Physics-Guided Vision Transformer Network With Tokens Complementarity for Hyperspectral Unmixing

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***Abstract*—Effectively modeling spatial–spectral information and local–global features is crucial for hyperspectral unmixing (HU). Some existing deep learning (DL)-based unmixing methods still exhibit limitations in global spectral modeling and physical guidance, which affect both the accuracy and interpretability of unmixing. To address these issues, in this article, we propose a physics-guided encoder (PGE) module, which employs a convolutional neural network (CNN) to generate feature representations for the image of each spectral band and incorporates the linear mixing model (LMM) to ensure physical consistency. Subsequently, these features are transformed into input tokens for a transformer, enabling spectral global modeling in the representations. Based on the PGE, we propose a physics-guided vision transformer network (PGViT-Net), which eliminates the need for manually adjusting variable shapes and ensures physical guidance in feature transformation. Furthermore, by integrating more spatial information tokens, we introduce a PGViT-Net with tokens complementarity (PGViT-TC-Net). This facilitates the joint modeling of spectral–spatial information and local–global feature representations, enhancing unmixing performance. Experiments on synthetic and real-world datasets demonstrate that our approach consistently overcomes state-of-the-art methods.**

***Index Terms*—Complementary tokens, hyperspectral unmixing (HU), local–global features, spatial–spectral information, vision transformer.**

NOMENCLATURE

$n = h \times w$	Number of pixels in hyperspectral image (HSI).
l	Number of bands in HSI.
r	Number of endmembers.
$\mathcal{Y} \in \mathbb{R}^{h\times w\times l}$	HSI.
$\mathbf{Y} \in \mathbb{R}^{l\times n}$	Mode-3 unfolded HSI.
$\mathbf{E} \in \mathbb{R}^{l\times r}$	Endmember matrix.
$\mathcal{A} \in \mathbb{R}^{h\times w\times r}$	Abundance tensor.
$\mathbf{A} \in \mathbb{R}^{r\times n}$	Abundance matrix.
$\mathcal{N} \in \mathbb{R}^{h\times w\times l}$	Noise in HSI.
$\mathbf{N} \in \mathbb{R}^{l\times n}$	Mode-3 unfolded noise.
$h_1 \times w_1$	Size of the output feature of layer 2 in CNN_1.
$h_2 \times w_2$	Size of the output feature of layer 3 in CNN_1.
l_1	Channel out of layer 1 in CNN_2.
l_2	Channel out of layer 2 in CNN_2.
l_3	Channel out of layer 3 in CNN_2.
$p \times p$	Size of the partitioned patch.
$\mathcal{Y}_1 \in \mathbb{R}^{h_2\times w_2\times r\times l}$	Token generated by CNN_1.
$\mathcal{Y}_2 \in \mathbb{R}^{h\times w\times l_3}$	Token generated by CNN_2.
$\mathbf{T}_1 \in \mathbb{R}^{l\times(h_2\cdot w_2\cdot r)}$	Input of transformer_1.
$\mathbf{T}_2 \in \mathbb{R}^{\left(\frac{h}{p}\cdot\frac{w}{p}\right)\times(p\cdot p\cdot l_3)}$	Input of transformer_2.
$\mathbf{T}_{t_1} \in \mathbb{R}^{1\times(h_2\cdot w_2\cdot r)}$	Output of transformer_1.
$\mathbf{T}_{t_2} \in \mathbb{R}^{1\times(p\cdot p\cdot l_3)}$	Output of transformer_2.

Received 9 January 2026; revised 14 February 2026; accepted 18 March 2026. Date of publication 23 March 2026; date of current version 2 April 2026. This work was supported in part by the National Key R&D Program of China under Grant 2025YFA1016400, in part by NSFC under Grant 12371517, and in part by Sichuan Science and Technology Program under Grant 2024NSFSC0038. *(Corresponding authors: Ting-Zhu Huang; Jie Huang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3676520

I. INTRODUCTION

HYPERSPECTRAL images (HSIs), compared to ordinary images, contain rich spectral information, making them widely used in various fields such as environmental monitoring, agricultural inspection, and mineral exploration [1], [2], [3], [4], [5], [6]. However, during data acquisition, the high-altitude, long-distance imaging of hyperspectral sensors, the limited spatial resolution of the instruments, and the effects of atmospheric scattering and absorption often result in data containing mixed pixels and noise. The mixed pixels refer to cases where multiple materials exist in a pixel. These factors increase the difficulty of applying HSIs. Therefore, hyperspectral unmixing (HU) is proposed to address this challenge, aiming to decompose mixed pixels into pure materials (endmembers) and their corresponding proportions (abundances) [7], [8]. Typically, the composition of endmembers in pixels is considered as linear and nonlinear mixing [7], [9], [10], [11]. The linear mixing models (LMMs) assume that light interacts with only one material before reaching the sensor [7], [12], [13]. The nonlinear mixing models (NLMMs) consider multiple interactions, such as scattering or reflections, among different materials, capturing more complex and realistic mixing effects [14], [15], [16]. For simplicity and physical interpretability, this article focuses on the LMM.

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Under the assumption of the LMM, many traditional models are proposed for HU. When spectral libraries are unavailable, both endmembers and abundances need to be estimated. This unmixing problem is regarded as a blind source separation (BSS) problem [17]. If pure endmembers exist in the HSIs, the vertex component analysis (VCA) is typically used first to extract these endmembers [18]. Subsequently, the fully constrained least squares (FCLS) is employed to estimate the corresponding abundances [19]. However, in real-world scenarios, pure endmembers may not always be present in the HSIs, leading to the above methods obtaining suboptimal performance. To address this limitation, nonnegative matrix factorization (NMF) is introduced into HU [20]. The principle of the NMF is to decompose the HSI matrix into two smaller matrices, which represent the endmembers and abundances, respectively. However, this decomposition problem is nonconvex and exists in many local minima. To address this issue, several additional constraints are introduced into NMF [21], [22], [23], [24]. For instance, the L_p norm is incorporated to impose sparsity constraints on the abundance, based on the assumption that each pixel contains only a subset of the endmembers, leading to many zero values in the abundance [25], [26], [27], [28]. Furthermore, the nuclear norm is introduced to enforce a low-rank constraint on the abundance matrix, accounting for the spatial correlations between pixels and indicating the linear dependencies between abundance vectors [13], [29], [30], [31]. Meanwhile, considering the intrinsic manifold structure of HSIs, graph regularization is employed to strengthen the relationship between the HSI and the related abundance [32]. It is noted that NMF first unfolds the 3-D HSI into a 2-D matrix. This process inevitably leads to the loss of spatial information. To overcome this limitation, matrix-vector nonnegative tensor factorization (MV-NTF) is proposed for HU, where the primary idea is to represent the 3-D HSI as the sum of the outer products of endmember curves and abundance maps to avoid the unfolding processing [33]. Furthermore, considering the piecewise smoothness of the abundance maps, the total variation (TV) regularization is introduced into NTF to improve the HU performance [34]. To enforce the low-rank structure of the abundance map more effectively, different from the simple decomposition applied in MV-NTF, the weighted nuclear norm regularization is introduced to constrain the low-rank property [35], [36]. Further considering the low-rank property of the abundance tensor formed by stacking abundance maps, references [37], [38] apply the weighted nuclear norm constraint on the abundance tensor to improve unmixing results.

In recent years, deep learning (DL) has demonstrated significant advantages in image processing, leading to its widespread application in HU. In particular, autoencoder (AE) based networks are extensively used in HU [39], [40], [41], [42], [43], [44], [45]. The AE network consists of two main components: an encoder and a decoder. The encoder maps the input high-dimensional data to a lower-dimensional embedding, while the decoder reconstructs this lower-dimensional embedding back into the high-dimensional space. In HU, the HSI is regarded as high-dimensional data, the abundance tensor as a low-dimensional embedding, and the weights of a single-layer convolutional decoder are considered as the endmembers. To avoid inaccurate or meaningless results, different encoder and decoder architectures are typically designed from various perspectives to enhance unmixing accuracy. For instance, it is noted that the nonnegative and tied encoder–decoder structures cannot effectively address the HU problem, while also recognizing that traditional denoising methods may introduce additional reconstruction errors (REs). Therefore, in [46], an untied denoising AE with sparsity (uDAS) has been proposed for HU. In [47], a stacked nonnegative sparse AE (NNSAE) has been presented for processing HSIs containing outliers and low signal-to-noise ratios (SNRs). The method is divided into two steps: the first step involves using a series of NNSAE to detect outliers in the data, and the second step extracts endmembers and abundances. In [48], an unsupervised two-stream AE network (TANet) has been proposed to address the limitations of previous methods in extracting physically meaningful endmembers. TANet adopts a two-stream architecture. First, superpixel segmentation is used to extract the endmember bundles from HSIs. Then, one stream learns the mapping from pseudopure pixels to abundances, while the other optimizes unmixing through an untied-weighted AE to minimize REs. AE-based network structures leverage their encoding–decoding mechanism to effectively learn mixed pixel features. However, their local feature extraction capability remains limited.

To model the spatial relationships between pixels and their neighboring pixels, convolutional neural networks (CNNs) have been introduced into HU tasks [49], [50], [51], [52], [53]. CNNs enhance feature extraction by integrating spatial information through convolutional operations. For example, a two-stream Siamese deep network, named endmember-guided unmixing network (EGU-Net), has been proposed for HU in [49]. This method learns an additional network from pure or near-pure endmembers to refine the weights of another unmixing network. The framework enhances the accuracy and stability of unmixing by sharing network parameters and incorporating spectrally meaningful constraints. This framework is not limited to per-pixel spectral unmixing but also integrates convolutional operators for spatial information modeling, enabling spatial–spectral unmixing. In [50], a multistage convolutional AE network (MSNet) has been presented to optimize the receptive field, enhancing contextual information modeling while preserving fine details, thereby improving the robustness and accuracy of unmixing in ill-posed scenarios. However, due to the limitation of the convolutional kernel size, CNNs can only extract local features, which significantly restricts their performance in HU tasks.

To address this issue, transformer, with its strong capability to model global dependencies, has been introduced into HU [54], [55], [56], [57], [58], [59]. An unmixing network, called DeepTrans, integrates CNN and transformer [54]. In particular, CNN is employed to encode the input HSI data, and then the transformer captures long-range dependencies among the representations extracted by the encoder. Finally, a convolutional decoder reconstructs the data. The spatial–spectral linear transformer unmixing network (SSLT-Net) combines grouped multiscale convolution with a linear transformer to extract

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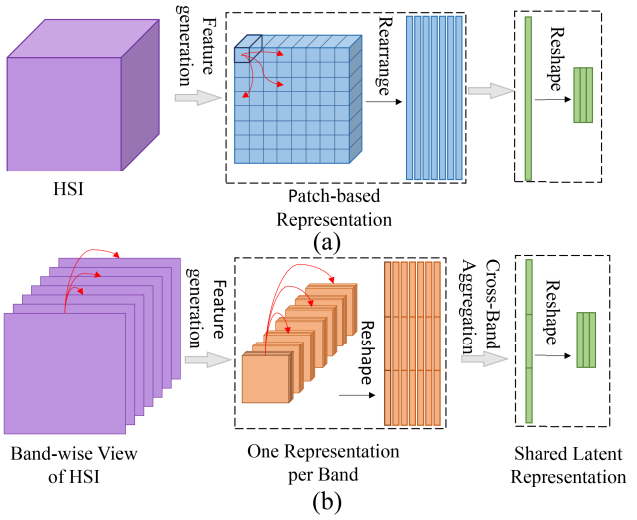


Fig. 1. Different encoding processes. (a) Conventional encoder. (b) Unmixing-process-guided encoder.

and fuse local, global, spectral, and spatial features through a dual-branch architecture, emphasizing feature diversity [55]. In addition, the endmember-oriented transformer network (EOT-Net) introduces endmember-specific attention mechanisms and heterogeneous feature fusion strategies to model spectral variability through endmember-oriented representations [59]. The transformer and CNN collaborative unmixing network (TCCU-Net) explores a dual-stream collaborative framework in which a transformer encoder and a pyramid CNN encoder are jointly designed to learn spectral, spatial, global, and local features, and their outputs are fused to enhance abundance estimation [52]. The multiscale aggregation transformer network (MAT-Net) in [58] addresses the limitations of a single receptive field by using a dual-stream, multibranch CNN encoder to extract spectral and multiscale spatial features, while a multihead self-scale aggregation attention (MSsaA) transformer encoder effectively integrates them. In addition, in [57], a U-shaped transformer network using shifted windows (UST-Net) has been proposed to address the loss of multiscale and spatial information. The network uses a U-shaped architecture with downsampling and upsampling stages to capture multiscale features and incorporates a shifted-window multihead self-attention mechanism to enhance spatial information, improving unmixing accuracy, stability, and spatial precision. Despite the promising performance achieved by these CNN–transformer hybrid architectures, most existing methods primarily focus on enhancing spectral–spatial feature interaction or multiscale contextual modeling from a data-driven perspective. In these approaches, transformer tokens are typically constructed from spatial patches or latent feature maps, and attention mechanisms are employed to improve feature complementarity. However, the physical interpretation of the token construction process is often weakly constrained. In particular, as shown in Fig. 1(a), spectral dimensions are frequently compressed or entangled during encoding without explicitly respecting the bandwise linear mixing property of HSIs, which may lead to the loss of physically meaningful spectral information and reduced interpretability of the unmixing results. Moreover, while global information modeling is commonly performed along the spatial dimension to capture long-range spatial dependencies, global spectral modeling remains relatively underexplored. As shown in Fig. 1(b), HSIs exhibit strong spectral correlations governed by the LMM, where each spectral band can be regarded as a linear projection of a shared set of abundance maps under different spectral coefficients. Relying solely on local convolutional operations or spatially constructed tokens may fail to fully capture such global spectral characteristics, thereby limiting unmixing accuracy. Therefore, designing an encoder that explicitly incorporates physical guidance during feature encoding while enabling effective global spectral modeling remains a significant challenge.

To address the aforementioned challenges, this article proposes a novel encoder module termed the physics-guided encoder (PGE). Unlike conventional CNN-based encoders that reduce the spectral dimension of HSIs to obtain compact latent representations, the proposed PGE is explicitly designed from the perspective of the LMM. In particular, instead of compressing spectral information across bands, the PGE employs a CNN to generate feature representations for each spectral band of the image. Under the LMM assumption, each band image is a linear combination of the same set of abundance maps with band-dependent spectral coefficients. Consequently, the feature representations extracted from different spectral bands should share a common underlying structure related to the abundance maps. Motivated by this physical interpretation, the PGE constructs bandwise feature representations that preserve the intrinsic linear mixing property of HSIs. These representations are then reshaped into bandwise tokens and fed into a transformer encoder. By modeling the global dependencies among these spectral tokens through self-attention, the transformer is encouraged to identify a shared representation corresponding to the common abundance-related basis across all spectral bands. In this manner, global spectral information is explicitly modeled in a physically interpretable way, rather than being implicitly learned through unconstrained feature compression. Based on the PGE, we proposed a named physics-guided vision transformer network (PGViT-Net) for HU, eliminating the need for manual adjustments in variable shapes, with the feature transformation process driven entirely by physical meaning. In addition, from the token perspective, the tokens generated by the PGE contain rich spectral information feature representation. Combining conventional token generation methods by introducing tokens containing abundant spatial information, and then fusing with spectral information through the transformer, we proposed a PGViT-Net with tokens complementarity (PGViT-TC-Net). This enables the output features to integrate both spectral–spatial and local–global information, significantly enhancing the unmixing performance. It is worth noting that the proposed PGE is explicitly constructed in its architectural design to preserve physically meaningful bandwise mixing structures. Meanwhile, the nonnegativity and sum-to-one constraints on abundances are enforced through the decoder architecture and corresponding activation functions. In particular, the main contributions of this article are as follows.

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1) We propose a PGE for HU. The PGE constructs band-wise feature representations and leverages the global dependency modeling capability of transformers to identify a shared abundance-related representation across spectral bands, enabling physically interpretable global spectral modeling.
2) Based on the PGE, we propose a PGViT-Net for HU. In this network, we minimize unnecessary modifications to the shape of the data. When modifications are necessary, we adjust the shape according to the physical significance of the variables, ensuring that the overall network process aligns more closely with physical principles.
3) Furthermore, to address the lack of global spatial dependency modeling in PGViT-Net, we introduce a token complementarity modeling mechanism and develop an extended network, termed PGViT-TC-Net. By explicitly exploiting the complementary relationships among physics-guided tokens, the proposed token complementarity module enables effective global spatial information aggregation without altering the core encoder architecture, thereby enhancing estimation performance.

The rest of this article is structured as follows. Section II briefly introduces the problem formulation related to HU and AE networks. Section III describes the detailed architecture of the proposed method. Section IV presents the experimental results. Finally, Section V concludes this article.

II. Background on Unmixing

For clarity, the main notations used throughout this article are summarized in Nomenclature. Given an HSI $\mathcal{Y} \in \mathbb{R}^{h\times w\times l}$ with $h \times w$ pixels and l bands, under the assumption of LMM, each pixel $\mathcal{Y}_{i,j,:}$ of the HSI is represented as

$$\mathcal{Y}_{i,j,:} = \mathcal{A}_{i,j,1} \times \mathbf{E}_{:,1} + \cdots + \mathcal{A}_{i,j,r} \times \mathbf{E}_{:,r} + \mathcal{N}_{i,j,:} \tag{1}$$

where $i \in \{1,\ldots,h\}$, $j \in \{1,\ldots,w\}$, $\mathbf{E} \in \mathbb{R}^{l\times r}$ represents the endmembers containing r signatures, $\mathcal{A} \in \mathbb{R}^{h\times w\times r}$ is formed by stacking abundance maps called the abundance tensor in [37], [38], and $\mathcal{N} \in \mathbb{R}^{h\times w\times l}$ denotes the noise. Furthermore, $\mathcal{Y}$ can be written as

$$\mathcal{Y} = \mathcal{A} \times_3 \mathbf{E} + \mathcal{N} \tag{2}$$

where $(\mathcal{A} \times_3 \mathbf{E})_{i,j,k} = \sum_{k'=1}^{r} \mathcal{A}_{i,j,k'}\mathbf{E}_{k,k'}$, for $i \in \{1,\ldots,h\}$, $j \in \{1,\ldots,w\}$, and $k \in \{1,\ldots,l\}$. In matrix form, we have

$$\mathbf{Y} = \mathbf{E}\mathbf{A} + \mathbf{N} \tag{3}$$

where $\mathbf{Y} \in \mathbb{R}^{l\times n}$, $\mathbf{A} \in \mathbb{R}^{r\times n}$, and $\mathbf{N} \in \mathbb{R}^{l\times n}$ are the mode-3 unfoldings of $\mathcal{Y}$, $\mathcal{A}$ and $\mathcal{N}$, respectively, and $n = h \times w$. Generally, the abundance maps need to satisfy two physical constraints, the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC)

$$\mathcal{A}_{i,j,k} \geq 0, \quad \sum_{k=1}^{r} \mathcal{A}_{i,j,k} = 1. \tag{4}$$

Recent DL-based unmixing methods achieve outstanding performance, with many solutions exploiting AEs. AE-based unmixing networks primarily consist of two components: an encoder and a decoder. The encoder analyzes the input HSI $\mathcal{Y}$ into an abundance tensor $\mathcal{A}$. Then, a single convolutional layer decoder converts the abundance tensor $\mathcal{A}$ to the reconstructed HSI $\hat{\mathcal{Y}}$. The weights of the layer are considered as endmembers in $\mathbf{E}$. Thus, we have

$$\mathcal{A} = f(\mathcal{Y}), \quad \hat{\mathcal{Y}} = \hat{f}(\mathcal{A}) \tag{5}$$

where $f : \mathbb{R}^{h\times w\times l} \to \mathbb{R}^{h\times w\times r}$ and $\hat{f} : \mathbb{R}^{h\times w\times r} \to \mathbb{R}^{h\times w\times l}$ are denoted as the encoder and the decoder, respectively.

TABLE I
Structure of the Proposed CNN_1. BN Stands for Batch Normalization

Layers	Layer 1	Layer 2	Layer 3
Composition	Conv 2D BN Dropout Leaky ReLU	Conv 2D BN Leaky ReLU	Conv 2D BN
Kernel	3×3		
Stride	1	2	2
Channels in	1	r	r
Channels out	r	r	r

III. Proposed Methodology

A. PGViT-Net

We usually focus on a single pixel $\mathcal{Y}_{i,j,:}$ in the HSI $\mathcal{Y}$, which is formed by a linear combination of r endmembers $\mathbf{E}$. Instead, if we focus on an image $\mathcal{Y}_{:,:,k}$ in a spectral band of the HSI $\mathcal{Y}$, it can be viewed as a linear combination of r abundance maps $\mathcal{A}$, i.e.,

$$\mathcal{Y}_{:,:,k} = \mathbf{E}_{k,1} \times \mathcal{A}_{:,:,1} + \cdots + \mathbf{E}_{k,r} \times \mathcal{A}_{:,:,r} \tag{6}$$

where $k \in \{1,\ldots,l\}$. By combining AE-based networks, we can interpret the encoder process as searching for a set of representatives that can be used to represent each spectral band in HSI. Based on this physical interpretation, we design PGViT-Net for HU.

The structure of our PGViT-Net is shown in Fig. 2. First, we use CNNs to generate a feature representation of an image for each band of the input HSI $\mathcal{Y}$. We denote it as CNN_1 [i.e., f_{CNN_1} in the following equation], and its structure is shown in Table I. This process is as follows:

$$\mathcal{Y}_1 = f_{\text{CNN}_1}(\mathcal{Y}) \tag{7}$$

where $\mathcal{Y} \in \mathbb{R}^{h\times w\times 1\times l}$ is the original HSI with one dimension expanding and $\mathcal{Y}_1 \in \mathbb{R}^{h_2\times w_2\times r\times l}$ is the generated feature representation. Noting that in (6), $\mathcal{A}_{:,:,1},\ldots,\mathcal{A}_{:,:,r}$ represent a common set of abundances for all $\mathcal{Y}_{:,:,k}$. For $\mathcal{Y}_1$, which contains l groups of information representations $\mathcal{Y}_{1_{:,:,k}} \in \mathbb{R}^{h_2\times w_2\times r\times 1}$, we leverage the global dependency modeling capability of the transformer to identify a representative set from these groups. To achieve it, we convert $\mathcal{Y}_1$ into tokens $\mathbf{T}_1 \in \mathbb{R}^{n_1\times d_1}$, where $n_1 = l$ is the numbers of tokens and $d_1 = h_2 \cdot w_2 \cdot r$ is the length of each token. In short, we generate features for each spectral band of the HSI, where each feature can be regarded as a representation of abundances. Each feature is then reshaped into a token, with each column forming a token,

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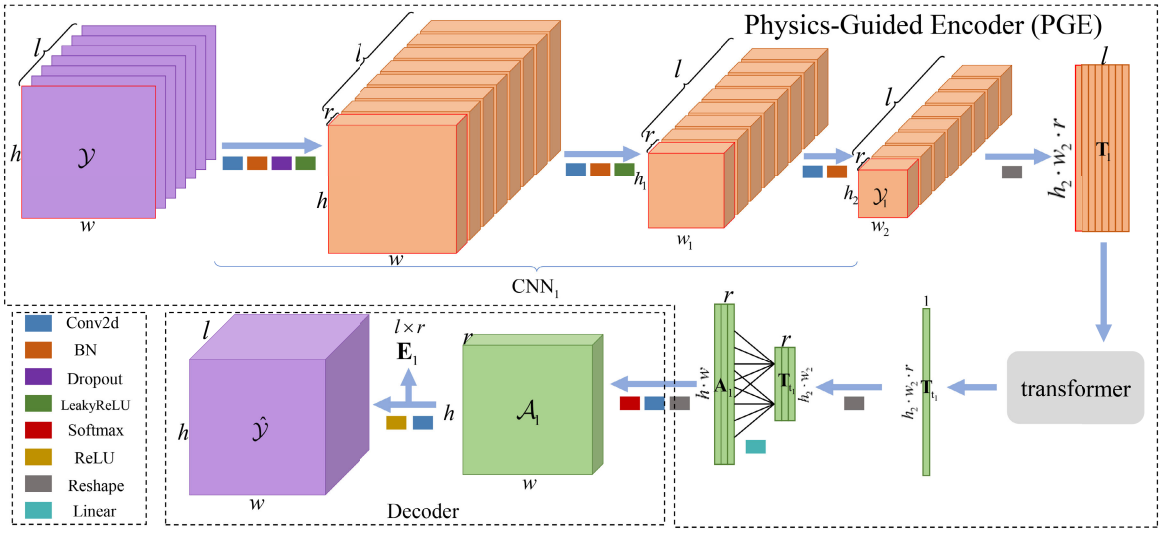


Fig. 2. Flowchart of the PGViT-Net. In the PGE module, the input HSI $\mathcal{Y}$ is first processed by CNN_1 to generate l groups of feature representations $\mathcal{Y}_1$, where each group corresponds to a physically interpretable latent component related to spectral unmixing. These feature groups are reshaped into bandwise tokens $\mathbf{T}_1$ and fed into a transformer to model global dependencies and shared representations across spectral bands. The transformer outputs $\mathbf{T}_{t_1}$ are then reshaped back into a spatial feature, followed by a fully connected layer with physically guided constraints to estimate the abundance maps $\mathcal{A}_1$. Finally, a single convolutional decoder reconstructs the HSI $\hat{\mathcal{Y}}$, where the convolutional kernel weights explicitly represent the estimated endmembers $\mathbf{E}_1$.

which serves as input to the transformer. Through this design, physical consistency across spectral bands is incorporated at the token level.

Next, according to [60], a learnable class token $\mathbf{T}_{1_{cls}} \in \mathbb{R}^{1\times d_1}$ and a positional storage token $\mathbf{T}_{1_{pos}} \in \mathbb{R}^{(1+n_1)\times d_1}$ are defined. Then, $\mathbf{T}_{1_{cls}}$ and $\mathbf{T}_1$ are concatenated and added to $\mathbf{T}_{1_{pos}}$ to generate $\mathbf{T}_1'$, i.e.,

$$\mathbf{T}_1' = \left(\mathbf{T}_{1_{cls}} \parallel \mathbf{T}_1\right) + \mathbf{T}_{1_{pos}} = \left(\mathbf{T}'_{1_{cls}} \parallel \mathbf{T}_1^*\right) \tag{8}$$

where $\mathbf{T}'_{1_{cls}} \in \mathbb{R}^{1\times d_1}$ and $\mathbf{T}_1^* \in \mathbb{R}^{n_1\times d_1}$ are new learnable class token and input tokens that both incorporate positional storage tokens, and $(\cdot \parallel \cdot)$ is the concatenation operator. Then, the multiple-head spectral-image attention (MSIA) is applied to $\mathbf{T}'_{1_{cls}}$ and $\mathbf{T}_1'$, which is mathematically expressed as

$$\text{MSIA}\left(\mathbf{T}_1'\right) = \mathbf{A}_1'\mathbf{v}_1' \tag{9}$$

where $\mathbf{A}_1' = \text{softmax}(\mathbf{q}_1'\mathbf{k}_1'^T/(d_1/m_{h1})^{1/2}) \in \mathbb{R}^{m_{h1}\times(n_1\cdot m_{h1})}$ is the attention weight, $\mathbf{q}_1' \in \mathbb{R}^{m_{h1}\times(d_1/m_{h1})}$, $\mathbf{k}_1' \in \mathbb{R}^{(n_1\cdot m_{h1})\times(d_1/m_{h1})}$, and $\mathbf{v}_1' \in \mathbb{R}^{(n_1\cdot m_{h1})\times(d_1/m_{h1})}$ are derived from the reshaped $\mathbf{q}_1 = \mathbf{T}'_{1_{cls}}\mathbf{W}_{\mathbf{q}_1} \in \mathbb{R}^{1\times d_1}$, $\mathbf{k}_1 = \mathbf{T}_1'\mathbf{W}_{\mathbf{k}_1} \in \mathbb{R}^{n_1\times d_1}$, and $\mathbf{v}_1 = \mathbf{T}_1'\mathbf{W}_{\mathbf{v}_1} \in \mathbb{R}^{n_1\times d_1}$, respectively, and are used to enhance the relationships by applying multiple heads. $\mathbf{W}_{\mathbf{q}_1}$, $\mathbf{W}_{\mathbf{k}_1}$, and $\mathbf{W}_{\mathbf{v}_1}$ are learnable parameters with the same size $d_1 \times d_1$. Then, the obtained $\text{MSIA}(\mathbf{T}_1')$ of size $m_{h1} \times (d_1/m_{h1})$ is reshaped to $1 \times d_1$ for the next step. The next step is to enhance the feature expression ability through a multilayered perceptron (MLP). Thus, we have

$$\mathbf{T}_1''' = \mathbf{T}_1'' + \text{MLP}\left(\text{LN}\left(\mathbf{T}_1''\right)\right) \tag{10}$$

where $\mathbf{T}_1'' = \mathbf{T}_{1_{cls}}'' \parallel \text{LN}(\mathbf{T}_1^*) \in \mathbb{R}^{(1+n_1)\times d_1}$, $\mathbf{T}_{1_{cls}}'' = \text{MSIA}(\mathbf{T}_1')\mathbf{W}_1 + \mathbf{T}'_{1_{cls}} \in \mathbb{R}^{1\times d_1}$, $\mathbf{W}_1 \in \mathbb{R}^{d_1\times d_1}$ are learnable weight parameters, and LN represents the layer normalization operation. Finally, we take the first column of $\mathbf{T}_1''' \in \mathbb{R}^{(1+n_1)\times d_1}$, the class token denoted as $\mathbf{T}_{t_1} = \mathbf{T}_{1_{\text{cls}}}''' \in \mathbb{R}^{1\times d_1}$, as output. Thus, we have

$$\mathbf{T}_{t_1} = f_{\text{transformer}_1}\left(\mathbf{T}_1\right) \tag{11}$$

where $f_{\text{transformer}_1} : \mathbb{R}^{l\times(h_2\cdot w_2\cdot r)} \to \mathbb{R}^{1\times(h_2\cdot w_2\cdot r)}$ fully utilizes its ability to capture long-distance dependencies to identify a set of information representations from the l groups. However, what we ultimately need are the abundance maps. Therefore, we still need to reshape the obtained $\mathbf{T}_{t_1} \in \mathbb{R}^{1\times(h_2\cdot w_2\cdot r)}$ to a size of $(h_2 \cdot w_2) \times r$. Then, after performing upsampling through the function $f_{\text{upsampling}}$, we obtain

$$\mathbf{A}_1 = f_{\text{upsampling}}\left(\mathbf{T}_{t_1}\right) \tag{12}$$

where $\mathbf{A}_1 \in \mathbb{R}^{(h\cdot w)\times r}$ is the abundance matrix, which is subsequently reshaped to obtain the abundance tensor $\mathcal{A}_1 \in \mathbb{R}^{h\times w\times r}$. Finally, the abundance tensor $\mathbf{A}_1$ is then transformed into the reconstructed HSI $\hat{\mathcal{Y}}$, through a single convolution layer

$$\hat{\mathcal{Y}} = \hat{f}\left(\mathcal{A}_1\right). \tag{13}$$

The weights of the convolution layer are the endmembers $\mathbf{E}$. We use the results of the VCA to initialize the weights of the decoder. Moreover, as illustrated in Fig. 2, physical constraints are enforced at the decoder stage, where Softmax ensures the nonnegativity and sum-to-one properties of the abundances, while ReLU guarantees nonnegative reconstruction HSI.

B. PGViT-TC-Net

It is worth noting that when generating the l group information representation, each group contains r pieces. When converting each group into a token, to prevent the length of a token from becoming excessively long, we reduce the output image size when using CNN_1 to generate each group's information. This operation results in the loss of some spatial detail information of $\mathcal{Y}$ within the tokens. To mitigate this

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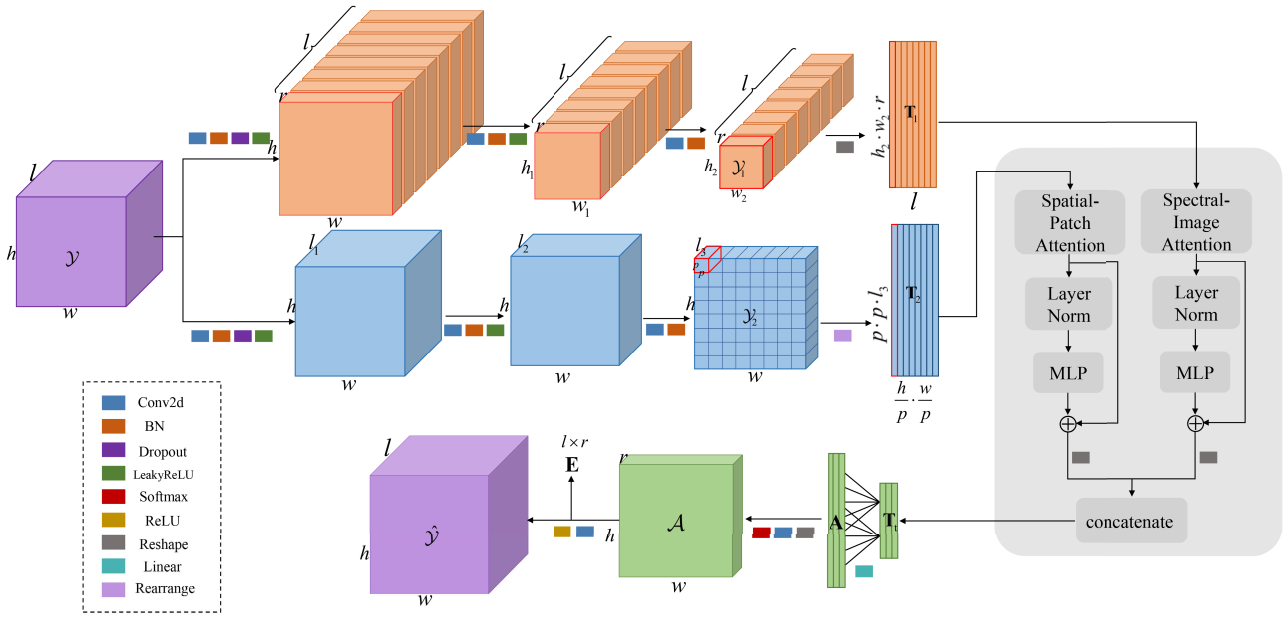


Fig. 3. Flowchart of our PGViT-TC-Net. In the encoder of PGViT-TC-Net, two complementary types of token representations are constructed. In particular, spectral-rich tokens $\mathbf{T}_1$ are generated by the PGE, which encodes physically interpretable bandwise information, while spatial-enhanced tokens $\mathbf{T}_2$ are obtained via a conventional patch-based feature generation strategy to capture local spatial structures, as illustrated in Fig. 4. These two token streams are separately fed into two transformer modules to model global dependencies along the spectral and spatial dimensions, respectively. The resulting features jointly encode spectral–spatial interactions as well as local–global contextual information. The fused representation $\mathbf{T}_t$ is then passed through a fully connected layer to estimate the abundance $\mathbf{A}$. Finally, a decoder reconstructs the HSI $\hat{\mathcal{Y}}$ from the estimated abundances $\mathcal{A}$.

TABLE II
STRUCTURE OF THE PROPOSED CNN_2

Layers	Layer 1	Layer 2	Layer 3
Composition	Conv 2D BN Dropout Leaky ReLU	Conv 2D BN Leaky ReLU	Conv 2D BN
Kernel	1×1		
Stride	1		
Channels in Channels out	l l_1	l_1 l_2	l_2 l_3

limitation, we propose a PGViT-TC-Net for HU, and its framework is illustrated in Fig. 3. We adopt the fusion concept and use another CNN to generate a second set of tokens, aiming to retain more spatial information. Hence, we have CNN_2, represented by f_{CNN_2}, whose structure is shown in Table II. Therefore, we have

$$\mathcal{Y}_2 = f_{\text{CNN}_2}(\mathcal{Y}) \tag{14}$$

where $\mathcal{Y}_2 \in \mathbb{R}^{h\times w\times l_3}$ and the output channel number l_3 is an adjustable parameter, which is set according to [54]. The values of channels out l_1 and l_2 of layers 1 and 2 in CNN_2 are fixed at 128 and 64, respectively. The output $\mathcal{Y}_2$ is then processed using a traditional patch-based method to generate $\mathbf{T}_2 \in \mathbb{R}^{(\frac{h}{p}\cdot\frac{w}{p})\times(p\cdot p\cdot l_3)}$ and the patch size is $p\times p$. In brief, features are generated by reducing the spectral dimensions, and then the spatial dimensions are partitioned into blocks, with each block treated as a feature. Each feature is then rearranged into a token by arranging its columns, forming the input to the transformer. In this way, we incorporate two complementary attention mechanisms: spectral-image attention and spatial-patch attention. Spectral-image attention is applied to tokens generated from individual spectral band images (i.e., $\mathbf{T}_1$ from CNN_1). It is designed to capture global spectral relationships under the LMM, where each spectral band can be viewed as a linear combination of the same set of abundance maps with different spectral coefficients. Spatial-patch attention, on the other hand, is applied to tokens generated from local spatial regions (i.e., $\mathbf{T}_2$ from CNN_2) and aims to capture long-range spatial dependencies within the HSI. Each token corresponds to a patch of the spatial feature map, enabling the model to learn spatial correlations across distant regions. By combining these two attention mechanisms, the resulting token representation $\mathbf{T}_t$ effectively integrates both spatial and spectral contextual information, providing richer cues for endmember and abundance estimation. According to [61], we use the transformer, denoted as $f_{\text{transformer}_2}$, to process these tokens yielding

$$\mathbf{T}_{t_2} = f_{\text{transformer}_2}(\mathbf{T}_2) \tag{15}$$

where $\mathbf{T}_{t_2} \in \mathbb{R}^{1\times(p\cdot p\cdot l_3)}$. To enable $\mathbf{A}$ to obtain more spatial information, we concatenate $\mathbf{T}_{t_1}$ and $\mathbf{T}_{t_2}$ to obtain $\mathbf{T}_t = \mathbf{T}_{t_1} \| \mathbf{T}_{t_2} \in \mathbb{R}^{(h_2\cdot w_2\cdot r+p\cdot p\cdot l_3)\times1}$. Then, we reshape $\mathbf{T}_t$ to a size of $(h_2\cdot w_2+\frac{p\cdot p\cdot l_3}{r})\times r$ and perform upsampling, through $f'_{\text{upsampling}}$, to obtain $\mathbf{A}$, that is,

$$\mathbf{A} = f'_{\text{upsampling}}(\mathbf{T}_t). \tag{16}$$

Finally, $\mathbf{A}$ is reshaped into $\mathcal{A}$, and then, passed through the decoder in (13) to obtain the reconstructed $\hat{\mathcal{Y}}$ and the endmembers $\mathbf{E}$.

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C. Loss Function

The loss function is divided into three components. Two losses focus on the reconstructed HSI $\hat{\mathcal{Y}}$ to effectively preserve spatial and spectral information. In particular, we adopt the RE loss and the spectral angle distance (SAD) loss, which are defined as

$$L_{\mathrm{RE}}\left(\mathcal{Y},\hat{\mathcal{Y}}\right)=\frac{1}{l}\sum_{k=1}^{l}\left\|\mathcal{Y}_{:,:,k}-\hat{\mathcal{Y}}_{:,:,k}\right\|_2^2$$

$$L_{\mathrm{SAD}}\left(\mathcal{Y},\hat{\mathcal{Y}}\right)=\frac{1}{h\times w}\sum_{i=1}^{h}\sum_{j=1}^{w}\arccos\frac{\left\langle\mathcal{Y}_{i,j,:},\hat{\mathcal{Y}}_{i,j,:}\right\rangle}{\left\|\mathcal{Y}_{i,j,:}\right\|_2\left\|\hat{\mathcal{Y}}_{i,j,:}\right\|_2} \tag{17}$$

respectively. $\|\cdot\|_2$, $\langle\cdot,\cdot\rangle$, and arccos are the ℓ_2 norm, the dot product, and the arccosine function, respectively. In addition, from [62], to optimize the endmembers, we introduce a loss, $L_{\mathbf{E}}$, to enforce the endmembers toward the center of the data simplex, defined as

$$L_{\mathbf{E}}=\left\|\mathbf{E}-\mathbf{m}\mathbf{1}_r^T\right\|_F^2 \tag{18}$$

where $\mathbf{m}\in\mathbb{R}^{l\times 1}$ are the mean values of the spectral pixels, $\|\cdot\|_F$ is the Frobenius norm, $\mathbf{1}_r$ is an all-ones vector of size r, and $\cdot^T$ is the transpose operator. Finally, the overall loss function, L, is formulated as

$$L=\beta L_{\mathrm{RE}}+\gamma L_{\mathrm{SAD}}+\theta L_{\mathbf{E}} \tag{19}$$

where β, γ, and θ serve as regularization parameters.

D. Complexity

In this section, we provide the computational complexity analysis of the proposed PGViT-Net and PGViT-TC-Net. First, we analyze PGViT-Net. In the encoding stage, features are extracted by a three-layer convolutional network. The computational complexity of each convolutional layer is given by $\mathcal{O}_{\mathrm{Conv1}}=\mathcal{O}(hwr)$, $\mathcal{O}_{\mathrm{Conv2}}=\mathcal{O}(h_1w_1r^2)$, and $\mathcal{O}_{\mathrm{Conv3}}=\mathcal{O}(h_2w_2r^2)$, where $h_1=\lfloor\frac{h-1}{2}\rfloor+1$, $w_1=\lfloor\frac{w-1}{2}\rfloor+1$, $h_2=\lfloor\frac{h_1-1}{2}\rfloor+1$, and $w_2=\lfloor\frac{w_1-1}{2}\rfloor+1$. Accordingly, the computational complexity of CNN_1 can be expressed as $\mathcal{O}_{\mathrm{CNN}_1}=\mathcal{O}(lh_1w_1r^2+lh_2w_2r^2)$. Subsequently, transformer$_1$ takes the spectral length $n_1=l$ as the number of tokens, and the feature dimension of each token is $d_1=h_2w_2r$. The linear projections for **qkv** and the MLP module both incur a computational complexity of $\mathcal{O}(n_1d_1^2)$, while the MSIA module introduces a computational cost of $\mathcal{O}(n_1d_1)$. Therefore, the overall computational complexity of transformer$_1$ is dominated by the quadratic term and can be written as $\mathcal{O}(l(h_2w_2r)^2)$. Considering that the upsampling and decoding stages only involve linear operations, the total computational complexity of PGViT-Net can be summarized as $\mathcal{O}_{\mathrm{PGViT\text{-}Net}}=\mathcal{O}(l(h_2w_2r)^2)$. For PGViT-TC-Net, the CNN_2 performs convolutional transformations on the features at the original spatial resolution. Its computational complexity is mainly dominated by the first convolution layer, which can be expressed as $\mathcal{O}(hwll_1)$. Subsequently, transformer$_2$ adopts a patch-based spatial token representation, where the number of tokens is $n_2=\frac{hw}{p^2}$ and the feature dimension is $d_2=p^2l_3$. As a result, the overall computational complexity of the attention and MLP modules is $\mathcal{O}(hwp^2l_3^2)$. By jointly considering the linear computational cost of the upsampling and decoding stages, the total computational complexity of PGViT-TC-Net can be formulated as $\mathcal{O}_{\mathrm{PGViT\text{-}TC\text{-}Net}}=\mathcal{O}(l(h_2w_2r)^2+hwll_1+hwp^2l_3^2)$.

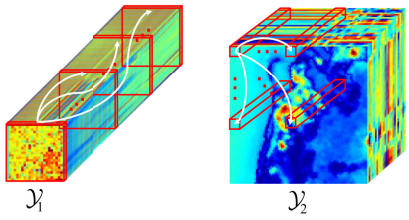


Fig. 4. Rich spatial feature representation and rich spectral feature representation are obtained by our PGViT-TC-Net on the Samson dataset, serving as an example.

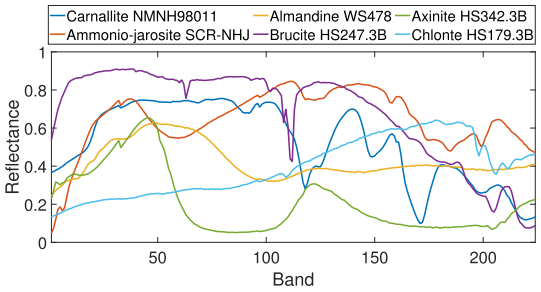


Fig. 5. Six spectral signatures from the USGS digital spectral library.

IV. Experiments

In the experiments, we compare the proposed networks with several advanced unmixing methods, including VCA-FCLS [18], [19], uDAS [46], UnDIP with endmembers generated by SiVM [63], MiSiC-Net [62], CyCU-Net [64], DeepTrans [54], A2SAN [65], UST-Net [57], MAT-Net [58], and EOT-Net [59]. We employ SAD and root mean square error (RMSE) to evaluate the performance of the estimated endmembers $\hat{\mathbf{E}}$ and abundances $\hat{\mathbf{A}}$ for all methods. They are defined as

$$\mathrm{SAD}_k=\arccos\frac{\mathbf{E}_k^T\hat{\mathbf{E}}_k}{\|\mathbf{E}_k\|_2\|\hat{\mathbf{E}}_k\|_2}$$

$$\mathrm{RMSE}_k=\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\mathbf{A}_{k,i}-\hat{\mathbf{A}}_{k,i}\right)^2} \tag{20}$$

respectively, where k indexes the kth endmember and abundance component, $\mathbf{E}$ and $\mathbf{A}$ are the reference endmembers and abundances and $\hat{\mathbf{E}}$, and $\hat{\mathbf{A}}$ denote the estimated endmembers and abundances. Lower SAD and RMSE values indicate better unmixing accuracy.

A. Hyperspectral Data

All methods are tested on two synthetic datasets and three real datasets.

1) *Synthetic Dataset 1 (SD1):* First, six signatures are chosen as endmembers from the USGS library, each containing 224 spectral bands. As shown in Fig. 5, these endmembers are Carnallite, Ammonio-jarosite, Almandine, Brucite, Axinite, and Chlonte. Second, the abundance maps are generated as follows. A 64 × 64

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TABLE III

QUANTITATIVE RESULTS USING THE SAD AND RMSE METRICS FOR THE COMPARED APPROACHES ON THE SD1. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

	Algorithms	VCA	uDAS	SiVM	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
SAD	End 1	0.06391	**0.03396**	0.08660	0.18987	0.15218	0.05614	0.16603	0.24435	0.06155	0.15949	0.07394	0.07725
	End 2	0.09949	0.10123	0.14702	0.22168	0.20722	0.10135	0.16340	0.13629	0.09878	0.17372	**0.09605**	0.10637
	End 3	0.13800	0.33158	0.17032	0.16581	0.20996	0.11473	0.31698	0.21646	0.13448	0.19260	0.12566	**0.09436**
	End 4	0.08716	0.07002	0.08635	0.15751	0.15903	0.10267	0.06766	0.08032	0.08386	0.13309	0.04199	**0.04152**
	End 5	0.29062	0.17763	0.41948	0.44258	0.38004	0.19478	0.22253	0.13939	0.27819	0.38887	0.18391	**0.13853**
	End 6	0.19085	0.34515	0.14672	0.30093	0.24986	0.12417	0.13628	0.18456	0.18110	0.20360	0.10705	**0.08822**
	Mean	0.14501	0.17659	0.17608	0.24640	0.22638	0.11564	0.17881	0.16690	0.13966	0.20856	0.10477	**0.09104**
	Algorithms	FCLS	uDAS	UnDIP	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
RMSE	End 1	0.09403	0.07509	0.12469	0.21036	0.21239	0.07973	0.11451	0.23166	0.07801	0.18061	0.09735	**0.05427**
	End 2	0.09723	0.05302	0.15021	0.18247	0.12152	0.08779	0.06736	0.16971	0.10580	0.22353	0.04920	**0.03813**
	End 3	0.14187	0.21782	0.17675	0.18791	0.31602	0.11629	0.20201	0.15572	0.13980	0.16820	0.09375	**0.06882**
	End 4	0.09651	0.05669	0.12490	0.32410	0.10953	0.07517	0.11641	0.21538	0.07828	0.16203	0.06726	**0.03740**
	End 5	0.10470	0.09872	0.09835	0.23319	0.14001	0.08529	0.13320	0.30298	0.11208	0.18387	0.07267	**0.06504**
	End 6	0.09860	0.16624	0.07640	0.15200	0.15008	0.06370	0.13293	0.27925	0.09542	0.14280	0.06459	**0.05141**
	Mean	0.10549	0.11126	0.12522	0.21500	0.17493	0.08466	0.12774	0.22578	0.10156	0.17684	0.07414	**0.05251**

synthetic image is divided into 64 equally sized square regions. Next, two endmembers are randomly selected to fill each region with proportions of 0.8 and 0.2, respectively. Then, a Gaussian blur is applied to increase the degree of mixing. After rescaling the fraction of each pixel to satisfy the ASC, the abundance is obtained. Finally, to obtain the SD1, Gaussian noise with a predefined SNR is added to the noise-free data, and the SNR is defined as

$$\text{SNR} = 10 \log_{10} \frac{\|\mathbf{Y}\|_F^2}{\|\mathbf{N}\|_F^2} \tag{21}$$

where $\mathbf{Y}$ is the synthetic dataset, $\mathbf{N}$ denotes the added noise, and $\log_{10}$ is the logarithm with base 10.

2) *Synthetic Dataset 2 (SD2):* This synthetic dataset contains 80×80 pixels and 200 spectral bands, covering a wavelength range from 1 to 2.5 μm [54]. It is assumed to contain three endmembers: CaO, SiO_2, and Fe_2O_3. Then, Gaussian noise is added with SNR = 30 dB, and impulse noise is introduced across all spectral bands, with each band undergoing perturbations affecting 0%–10% of its pixels. Therefore, the SD2 is obtained by the noise added, similar to [66].
3) *Samson Dataset:* It consists of 95×95 pixels and 156 spectral bands, spanning a wavelength range of 0.401 to 0.889 μm. It contains three endmembers—Soil, Tree, and Water—which are commonly used in unmixing [54].
4) *Apex Dataset:* This dataset is obtained by the APEX sensor, a cropped image with a size of 110×110 pixels and 285 bands, ranging from 0.413 to 2.420 μm. Four endmembers are considered: Water, Tree, Road, and Roof [54].
5) *Jasper Ridge Dataset:* The dataset contains 512×614 pixels with 224 spectral channels covering the wavelength range from 0.380 to 2.500 μm. To reduce the effects of strong water vapor absorption and atmospheric interference, a subimage of size 100×100 pixels is selected from the original data, and 198 effective spectral channels are retained. It is assumed that the selected subimage contains four endmembers, namely Tree, Water, Soil, and Road [57].

Table III presents the SAD and RMSE results of the estimated endmembers and abundances by all methods on the SD1 with SNR = 20 dB. Table III indicates that our PGViT-Net, benefiting from its network architecture guided by the physical mechanism of spectral unmixing, achieves the second-best performance in most cases for both endmember and abundance estimation. Based on our PGViT-Net, PGViT-TC-Net further enhances the integration of spectral and spatial features in token representations, leading to improved estimation performance. It achieves the best endmember estimation results except for Ends 1 and 2, as well as the best abundance estimation results for all endmembers. UST-Net and MAT-Net achieve the second-best SAD for End 5 and End 2, respectively. Fig. 6 illustrates the estimated endmember curves obtained by different methods. The endmember curves in Fig. 6 show that our PGViT-TC-Net provides estimates for Ends 3, 5, and 6 that are closer to the reference than other methods. The visual comparison Fig. 7 shows the estimated abundance maps. From Fig. 7, it is evident that abundance maps obtained by our PGViT-Net are more consistent with the reference for Ends 4–6. These results indicate that incorporating a physically guided network structure and optimizing token representations effectively enhances HU performance.

Table IV presents the estimation results of endmembers and abundances by all methods on the SD2. The results in Table IV show that our PGViT-Net and PGViT-TC-Net achieve the best and second-best performance in all endmember estimations, and also provide competitive outcomes in most abundance estimations. Moreover, the overall performance of both the proposed methods on the SD2 is relatively similar, which can be attributed to the block-like regular spatial distribution of the SD2. This pattern indeed limits the benefits gained from

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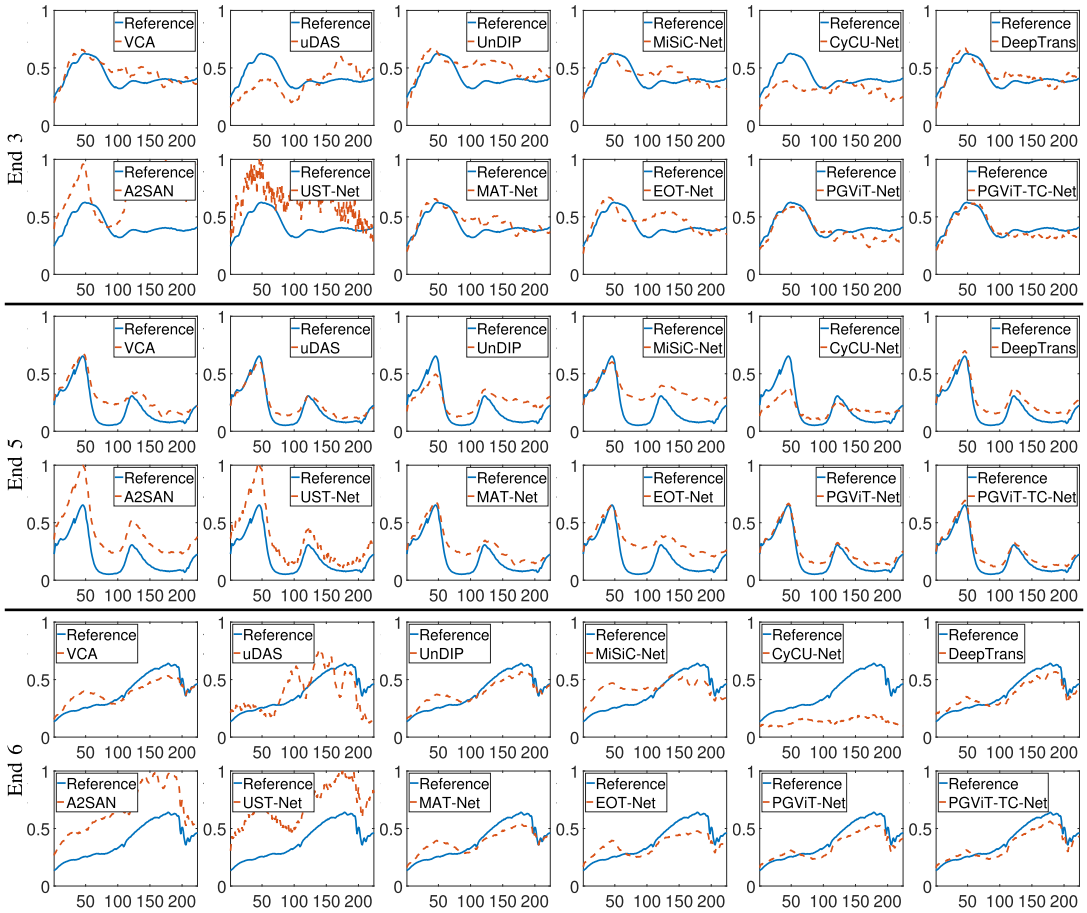


Fig. 6. Estimated endmember curves using the compared approaches on the SD1. The horizontal and vertical axes of each subfigure represent spectral bands and reflectance (%), respectively.

TABLE IV

COMPARISON OF SAD, RMSE, AND PSNR BY DIFFERENT METHODS ON THE SD2. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Algorithms		VCA	uDAS	SiVM	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
SAD	CaO	0.01392	0.01764	0.00666	0.02683	0.01971	0.01722	0.00494	0.01563	0.01226	0.00472	0.00439	**0.00431**
	SiO_2	0.00655	0.01325	0.00507	0.01095	0.01365	0.00734	0.00404	0.04274	0.00524	0.00594	0.00347	**0.00342**
	Fe_2O_3	0.02787	0.02584	0.02895	0.02249	0.02061	0.01933	0.00530	0.02601	0.02734	0.00549	**0.00430**	0.00450
	Mean	0.01611	0.01891	0.01356	0.02009	0.01799	0.01463	0.00476	0.02813	0.01494	0.00538	**0.00405**	0.00408
Algorithms		FCLS	uDAS	UnDIP	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
RMSE	CaO	0.04949	0.08799	0.05160	0.06883	0.38371	0.05212	0.05997	0.26431	0.09332	0.06988	**0.04076**	0.04087
	SiO_2	0.07812	0.11421	0.09180	0.09068	0.20208	0.07014	0.07190	0.22808	0.14099	0.09062	**0.05915**	0.05930
	Fe_2O_3	0.09009	0.09193	0.10720	0.10532	0.36596	0.07619	**0.04879**	0.15598	0.09564	0.08409	0.06893	0.06910
	Mean	0.07257	0.09804	0.08353	0.08828	0.31725	0.06615	0.06022	0.21612	0.10998	0.08153	**0.05628**	0.05642

incorporating additional spatial feature information. Fig. 8 shows the estimated endmember spectral curves on the SD2 for all methods. As illustrated in Fig. 8, our PGViT-Net and PGViT-TC-Net produce the most accurate estimations of endmembers for CaO and Fe_2O_3, while SiVM yields the most accurate result for SiO_2, compared to the reference. Fig. 9 illustrates the estimated abundance maps on the SD2. As shown in Fig. 9, the abundance maps generated by the proposed PGViT-Net and PGViT-TC-Net are the closest to the reference maps for all endmembers.

Table V presents the SAD and RMSE results of all methods on the Samson dataset. Table V reveals that our PGViT-Net achieves the second-best performance in the SAD of Tree and RMSE of Soil, as well as in the mean SAD. Our PGViT-TC-Net ranks second in the SAD of Soil and the RMSE of Tree and Water while achieving the best performance in the SAD of Tree and Water, mean SAD, the RMSE of Soil, and mean RMSE. CyCU-Net achieves the best result in the SAD of Soil, and MiSiC-Net obtains the lowest RMSE for Tree and Water. EOT-Net achieves the second-best SAD for

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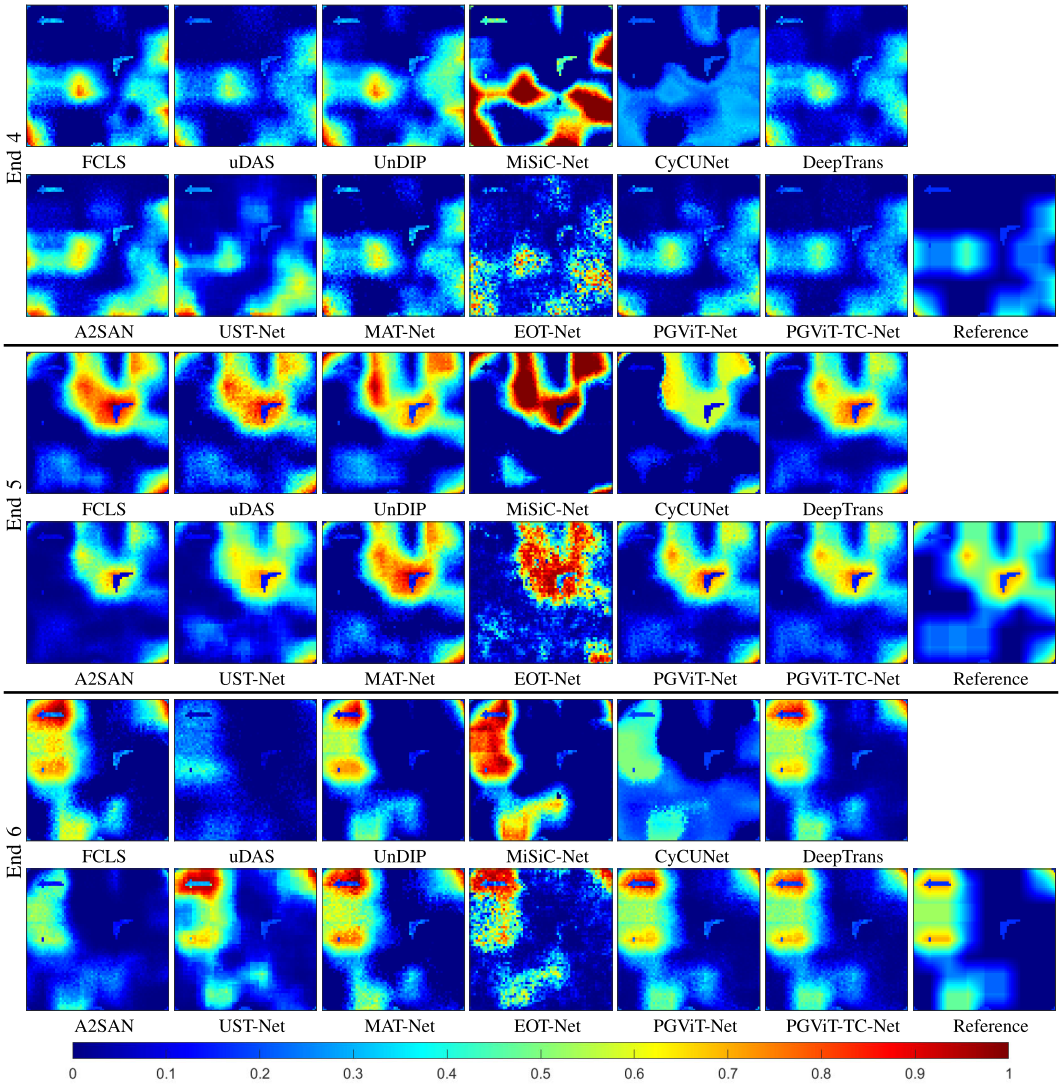


Fig. 7. References and estimated abundance maps provided by the compared approaches on the SD1.

TABLE V

COMPARISON OF SAD AND RMSE USING DIFFERENT METHODS ON THE SAMSON DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Algorithms		VCA	uDAS	SiVM	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
SAD	Soil	0.02592	0.03649	0.02592	0.02218	**0.01352**	0.02479	0.02612	0.01776	0.01944	0.01484	0.01745	0.01733
	Tree	0.09578	0.09724	0.07477	0.05223	0.05404	0.08323	0.05435	0.07215	0.09618	0.08014	0.04745	**0.04691**
	Water	0.15544	0.14572	0.15545	0.24015	0.19751	0.11465	0.04951	0.05280	0.05274	0.12296	0.05351	**0.04732**
	Mean	0.09238	0.09315	0.08538	0.10485	0.08836	0.07423	0.04333	0.04757	0.05612	0.07265	0.03947	**0.03718**
Algorithms		FCLS	uDAS	UnDIP	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
RMSE	Soil	0.17577	0.18069	0.17782	0.05716	0.07996	0.08109	0.15526	0.19503	0.14051	0.17655	0.04793	**0.04401**
	Tree	0.07722	0.13715	0.13306	**0.03465**	0.07540	0.05425	0.19455	0.12520	0.12598	0.13576	0.04017	0.03742
	Water	0.15880	0.22913	0.20957	**0.02923**	0.08271	0.05694	0.30333	0.25329	0.17272	0.24662	0.04094	0.03884
	Mean	0.13726	0.18232	0.17348	0.04035	0.07936	0.06409	0.21771	0.19117	0.14640	0.18631	0.04301	**0.04009**

Soil. Overall, the proposed PGViT-Net and PGViT-TC-Net demonstrate performance advantages across multiple evaluation metrics compared to the other methods. Fig. 10 shows the estimated endmembers obtained by all methods. It can be seen that CyCU-Net provides the most accurate estimation for Soil, followed by our PGViT-Net and PGViT-TC-Net. For Tree and Water, the proposed PGViT-Net and PGViT-TC-Net yield the closest spectral curves to the reference, while several other methods also perform well on Water. Fig. 11 displays the abundance maps estimated by all methods. As

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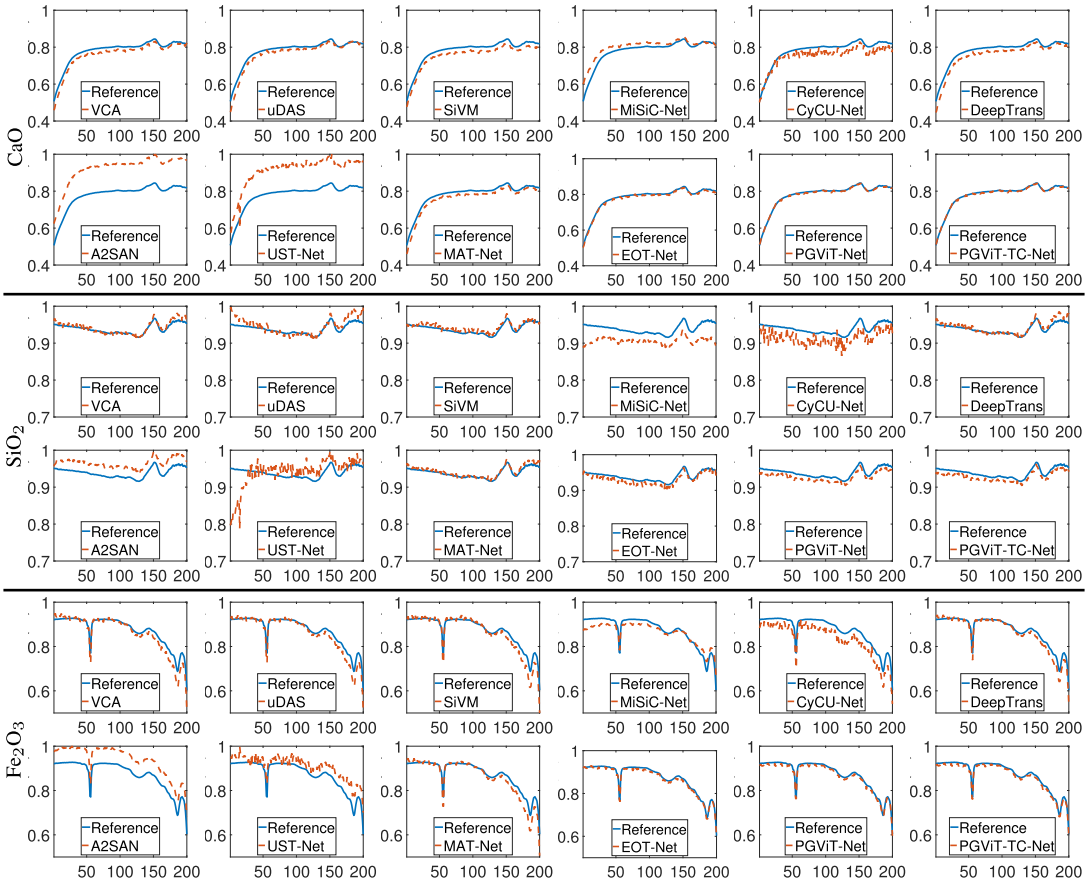


Fig. 8. Estimated endmember curves of the SD2 using different methods. The horizontal and vertical axes of each subfigure represent spectral bands and reflectance (%), respectively.

TABLE VI

COMPARISON OF SAD AND RMSE USING DIFFERENT METHODS ON THE APEX DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Algorithms		VCA	uDAS	SiVM	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
SAD	Road	0.69150	0.28964	0.81789	0.27108	0.33306	0.06777	0.06799	0.13750	0.72523	0.14312	0.08453	**0.06529**
	Tree	0.24905	**0.12374**	0.13289	0.13751	0.14932	0.13041	0.13252	0.15272	0.21082	0.17900	0.13009	0.13226
	Roof	0.12513	0.10383	1.17328	0.18968	0.16804	0.08562	0.08643	**0.05772**	0.11267	0.06106	0.13986	0.14210
	Water	0.51763	0.28337	0.49378	0.74526	0.19138	0.04934	**0.04163**	0.04477	0.22056	0.09768	0.04176	0.04464
	Mean	0.39583	0.20014	0.65446	0.33588	0.21045	0.08329	**0.08214**	0.09818	0.31732	0.12021	0.09906	0.09607
Algorithms		FCLS	uDAS	UnDIP	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
RMSE	Road	0.20656	0.18087	0.20401	0.13220	0.29645	0.17870	0.16824	0.14947	0.36044	0.12835	0.13592	**0.12507**
	Tree	0.14421	0.10817	0.25589	0.14590	0.19202	0.09479	0.13596	0.13831	0.23433	0.13445	0.08609	**0.07207**
	Roof	0.13428	0.22028	0.50024	**0.10584**	0.15942	0.13855	0.11541	0.11235	0.20495	0.15453	0.10994	0.10837
	Water	0.13366	0.31269	0.46760	0.18434	**0.06900**	0.08545	0.16899	0.14449	0.16175	0.21078	0.07537	0.07734
	Mean	0.15468	0.20550	0.35694	0.14207	0.17922	0.12437	0.14715	0.13615	0.24037	0.15703	0.10183	**0.09571**

shown in Fig. 11, our PGViT-Net and PGViT-TC-Net generate the most accurate abundance estimations for Soil and Water. For Tree, MiSiC-Net, DeepTrans, PGViT-Net, and PGViT-TC-Net all show competitive performance. Overall, the proposed PGViT-TC-Net achieves strong overall performance.

Table VI presents the SAD and RMSE of all methods on the Apex dataset. According to the results, our PGViT-Net achieves the second-best SAD and RMSE of the estimated Tree and Water, and mean RMSE. PGViT-TC-Net obtains the best results in SAD and RMSE of Road, as well as in RMSE of Tree and mean RMSE. UST-Net achieves the best estimated result for Roof. A2SAN achieves the lowest SAD for Water and the best mean SAD. Overall, the proposed PGViT-Net and PGViT-TC-Net achieve improved performance in abundance estimation. Fig. 12 illustrates the estimated endmembers of all methods on the Apex dataset. It is observed that our PGViT-Net and PGViT-TC-Net provide the best estimations for Road and Tree. CyCU-Net yields the most accurate

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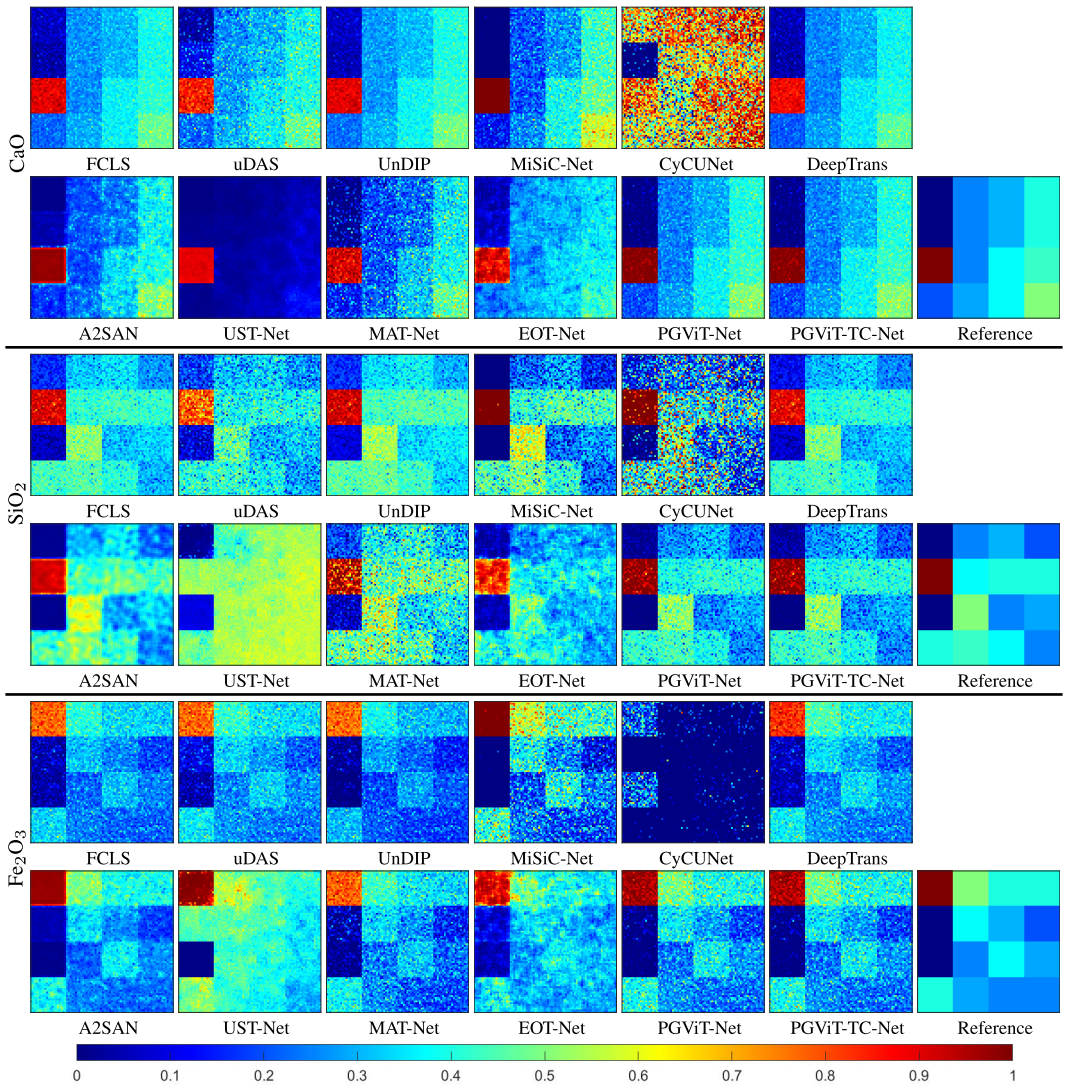


Fig. 9. References and estimated abundance maps of the SD2 using different methods.

TABLE VII

COMPARISON OF SAD AND RMSE USING DIFFERENT METHODS ON THE JASPER RIDGE DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

	Algorithms	VCA	uDAS	SiVM	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
SAD	Tree	0.10704	0.18991	0.10863	0.04348	0.14072	0.04786	0.07642	0.06604	0.10606	0.09277	0.02807	**0.02257**
	Water	0.24807	0.13296	0.29458	0.28985	0.17476	0.05979	0.08608	**0.02920**	0.09735	0.06493	0.05928	0.10786
	Soil	0.22341	0.11188	0.11995	0.17566	0.16852	0.03838	0.04428	**0.03771**	0.22326	0.03956	0.04245	0.05347
	Road	0.05366	0.06300	0.27548	0.16337	0.32661	0.07855	0.18411	0.04090	0.05183	0.03299	**0.01902**	0.02928
	Mean	0.15805	0.12444	0.19966	0.16809	0.20265	0.05615	0.09772	0.04346	0.11962	0.05756	**0.03720**	0.05330
	Algorithms	FCLS	uDAS	UnDIP	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
RMSE	Tree	0.15456	0.14080	0.15898	0.09569	0.15174	0.06463	0.08192	0.05822	0.10768	0.06883	**0.05092**	0.05460
	Water	0.12931	0.17511	0.16038	0.11437	0.09393	0.11095	0.08437	**0.03260**	0.07909	0.07684	0.07582	0.07496
	Soil	0.16702	0.13252	0.20420	0.34698	0.25430	0.10925	0.10530	0.08022	0.10407	0.11462	0.07397	**0.06667**
	Road	0.12707	0.12166	0.19217	0.34983	0.21696	0.12110	0.09460	0.06582	0.06502	0.08430	**0.06488**	0.06517
	Mean	0.14449	0.14252	0.17893	0.22672	0.17923	0.10149	0.09155	**0.05922**	0.08897	0.08615	0.06640	0.06535

estimation for Roof, while CyCU-Net and PGViT-TC-Net show a better performance in estimating Water. Fig. 13 shows the abundance maps estimated by all methods on the Apex dataset. From Fig. 13, PGViT-TC-Net produces abundance maps for Road and Tree that are closer to the references compared with other methods. For Roof, the results from

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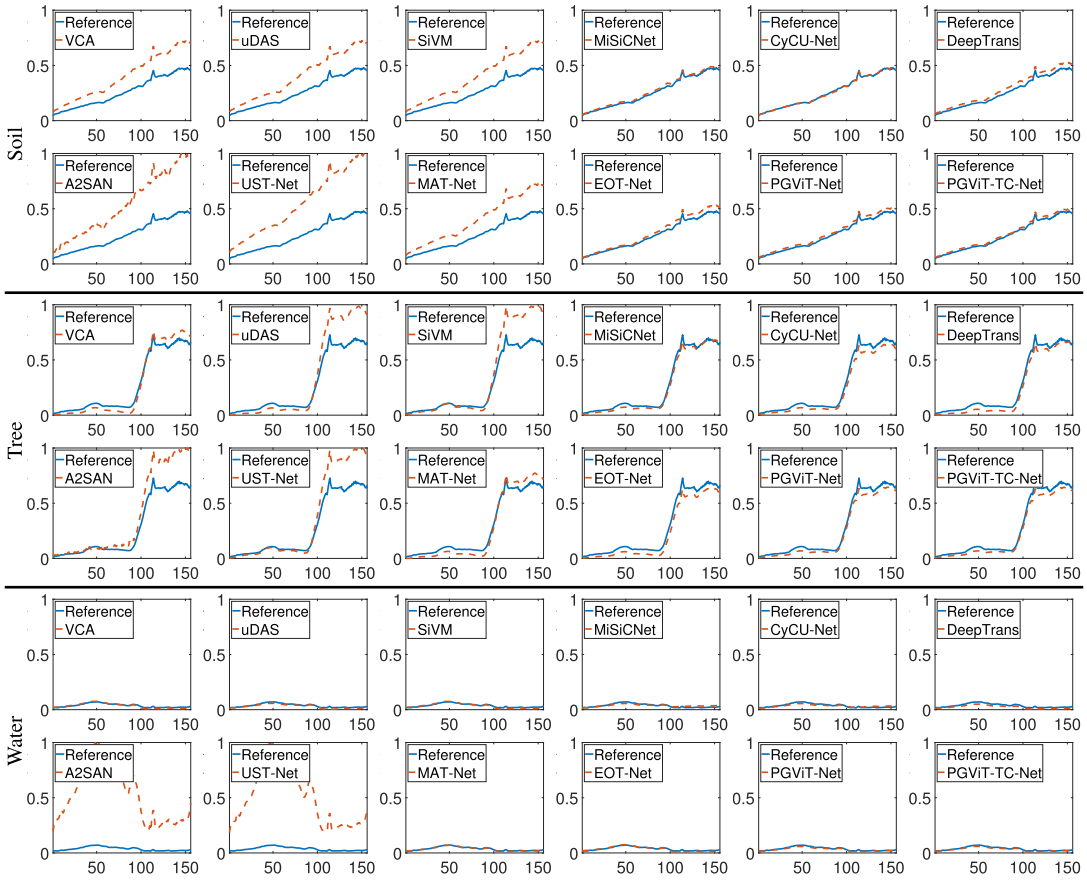


Fig. 10. Estimated endmember curves of the Samson dataset using different methods. The horizontal and vertical axes of each subfigure represent spectral bands and reflectance (%), respectively.

MiSiC-Net, PGViT-Net, and PGViT-TC-Net are closer to the reference. In terms of Water, PGViT-Net and PGViT-TC-Net achieve the best accurate estimations. Overall, our PGViT-Net and PGViT-TC-Net demonstrate better accuracy in abundance estimation on the Apex dataset.

Table VII shows the SAD and RMSE results of different unmixing methods on the Jasper ridge dataset. From the SAD, most deep-learning-based methods achieve clear improvements over traditional approaches, indicating the effectiveness of deep feature modeling in preserving spectral information. Among them, UST-Net and DeepTrans exhibit competitive performance on several endmembers. The proposed PGViT-Net attains the best results on the Road as well as the lowest mean SAD, while PGViT-TC-Net further improves the performance on the Tree. These results suggest that the proposed physics-guided spectral token aggregation strategy is beneficial for more accurate endmember characterization. In terms of the RMSE, UST-Net achieves the lowest average RMSE, demonstrating the effectiveness of its multiscale spatial modeling capability for abundance reconstruction. Meanwhile, PGViT-Net yields the best reconstruction accuracy on the Tree and Road, and PGViT-TC-Net achieves the best performance on Soil, indicating that the proposed methods maintain competitive abundance estimation performance across different materials. Overall, different methods exhibit distinct advantages in endmember estimation and abundance reconstruction. The proposed PGViT-Net and PGViT-TC-Net demonstrate strong overall stability and favorable average performance, supporting the effectiveness of integrating physics-guided spectral representations with the global modeling capability of transformers for HU. Fig. 14 illustrates the estimated endmembers obtained by all compared methods. It can be observed that EOT-Net provides the closest estimated endmembers to the references for Tree and Soil, highlighting the effectiveness of endmember-oriented guidance. In contrast, PGViT-Net and PGViT-TC-Net yield more accurate estimations for Water and Road, indicating that the proposed physics-guided transformer is capable of effectively capturing discriminative bandwise spectral information. Fig. 15 presents the abundance maps estimated by different methods. UST-Net achieves superior abundance estimation for Water and Soil, which can be attributed to its U-shaped architecture that facilitates multiscale information aggregation and global spatial modeling. Meanwhile, PGViT-Net and PGViT-TC-Net achieve better results on Tree and Road, highlighting the effectiveness of the proposed feature generation and fusion strategy in extracting task-relevant information.

B. *Influence of Noise on Unmixing*

To evaluate the impact of noise on the performance of our PGViT-Net and PGViT-TC-Net, both methods are evaluated

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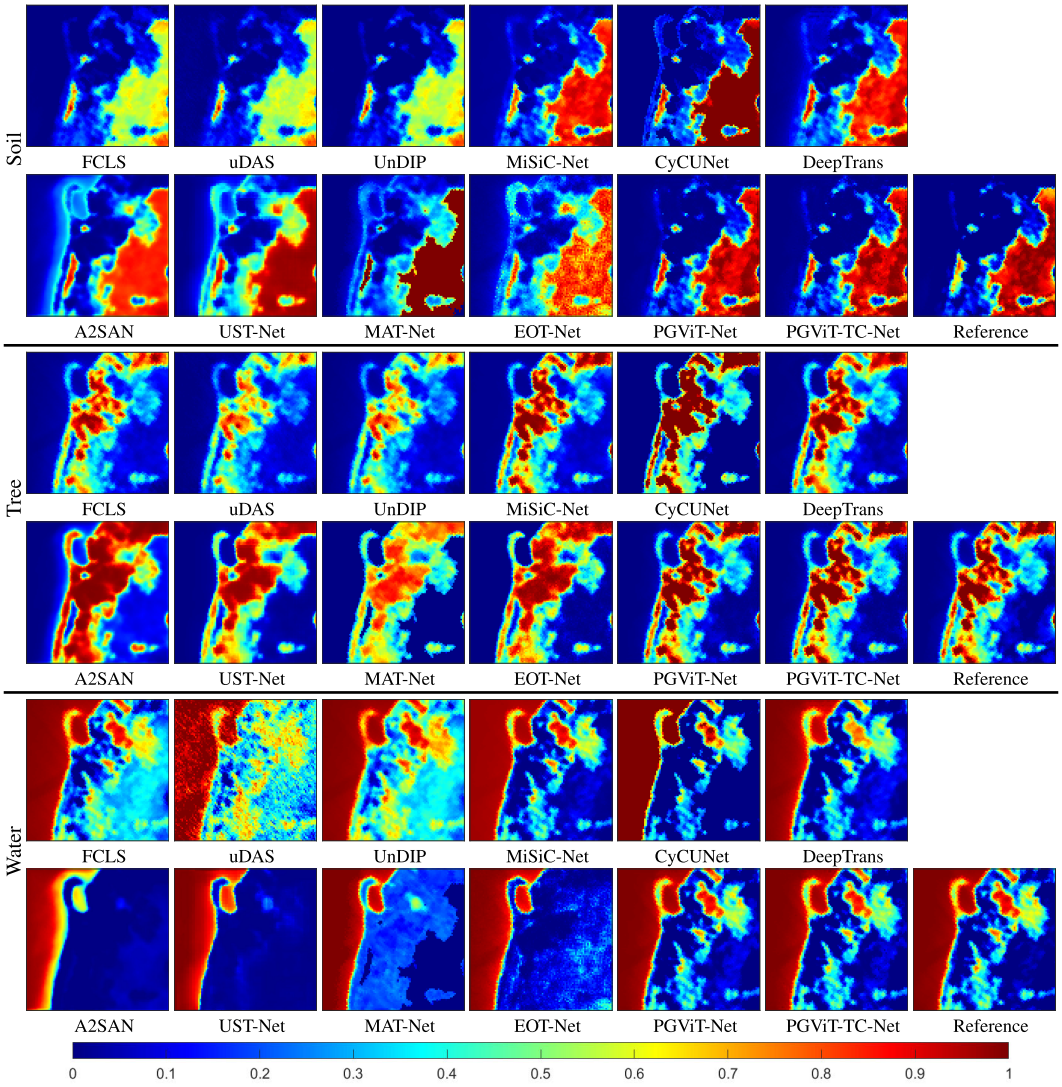


Fig. 11. References and estimated abundance maps of the Samson dataset using different methods.

on the SD1 under varying SNR levels of 10, 15, 20, 25, and 30 dB. The experimental results are shown in Fig. 16. As observed from Fig. 16, the SAD and RMSE for both estimated endmembers and abundances generally exhibit a decreasing trend as the SNR increases.

C. Sensitivity Analysis of Hyperparameters

In this section, we provide results about the hyperparameter selection strategy for PGViT-Net and PGViT-TC-Net. Both methods include the following hyperparameters: learning rate (LR), weight decay (WD), and the weighting coefficients β, γ, and θ used to balance different components of the loss function. To reduce the complexity of hyperparameter selection, θ is typically chosen from the set {1e−4, 1e−3, 1e−2}. The search spaces for the remaining hyperparameters are defined as follows.

1) LR $\in \{1e{-}3,\ 2e{-}3,\ \ldots,\ 8e{-}3,\ 9e{-}3\}$.
2) WD $\in \{1e{-}5,\ 2e{-}5,\ \ldots,\ 8e{-}5,\ 9e{-}5\}$.
3) $\beta \in \{1e{-}2,\ 5e{-}2,\ \ldots,\ 1e5,\ 5e5\}$.
4) $\gamma \in \{1e{-}4,\ 5e{-}4,\ \ldots,\ 1,\ 5\}$.

TABLE VIII
EMPIRICAL SET OF HYPERPARAMETERS

Hyperparameters		LR	WD	β	γ	θ
PGViT-Net	SD1	2e-3	8e-5	5e1	5	1e-3
	SD2	2e-3	2e-5	5e1	5e-4	1e-3
	Samson dataset	4e-3	4e-5	5e3	1e-3	1e-2
	Apex dataset	4e-3	4e-5	1e2	5e-4	1e-3
	Jasper ridge dataset	2e-3	2e-5	5e-1	1e-1	1e-3
PGViT-TC-Net	SD1	2e-3	9e-5	5e1	5	1e-4
	SD2	2e-3	2e-5	5e1	5e-4	1e-3
	Samson dataset	4e-3	4e-5	5e4	5e-3	1e-3
	Apex dataset	4e-3	4e-5	1e2	5e-4	1e-3
	Jasper ridge dataset	2e-3	2e-5	5e-1	5e-4	1e-3

We present the performance of PGViT-Net and PGViT-TC-Net varying LR, WD, β, and γ. As shown in Fig. 17, the values of SAD and RMSE across all datasets fluctuate by varying hyperparameter settings. From Fig. 17, it is observed that PGViT-Net and PGViT-TC-Net exhibit generally consistent

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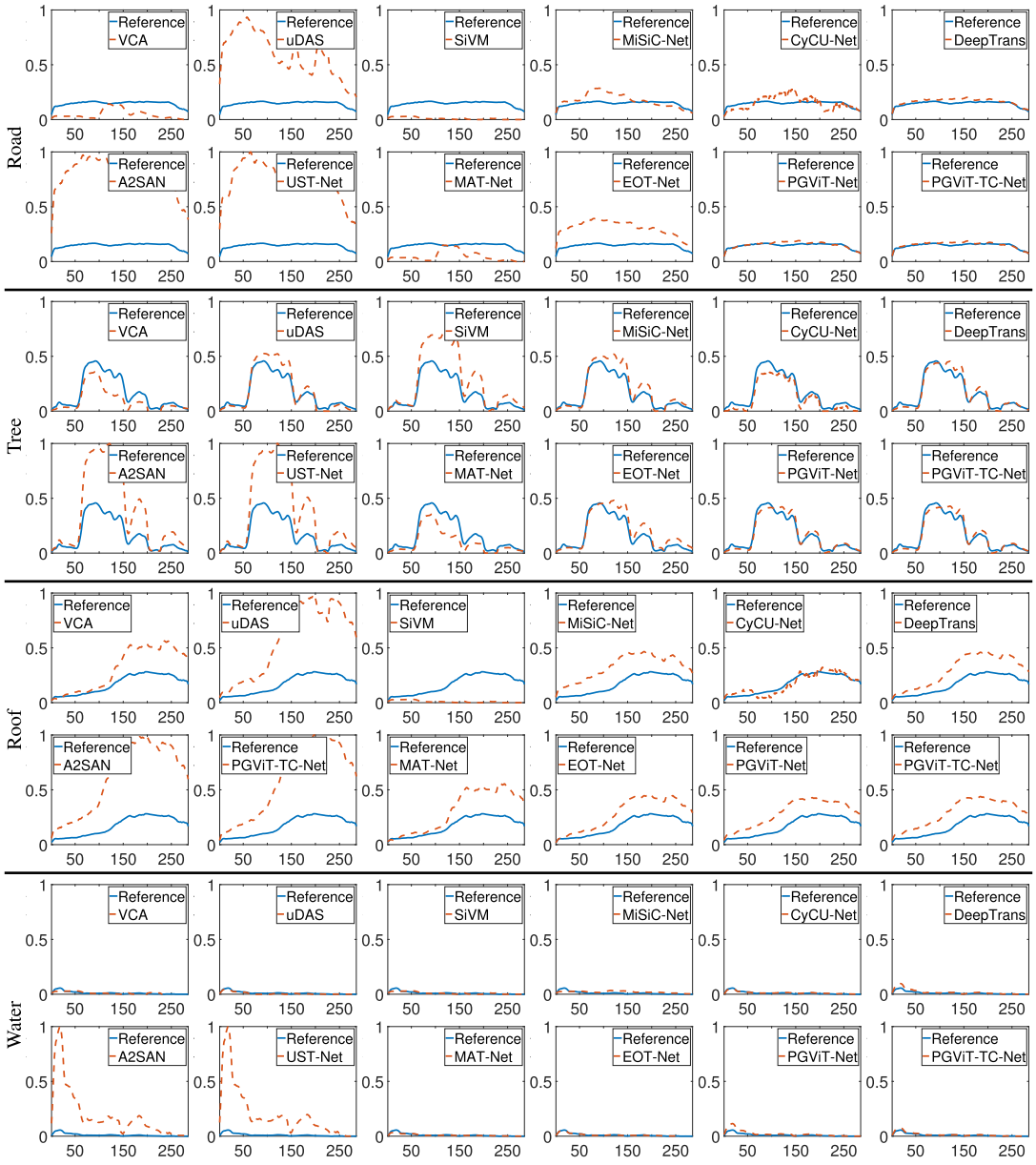


Fig. 12. Estimated endmember curves of the Apex dataset using different methods. The horizontal and vertical axes of each subfigure represent spectral bands and reflectance (%), respectively.

performance trends across different datasets under various parameter configurations, indicating that both methods share similar sensitivity characteristics with respect to hyperparameters. In particular, in terms of LR, optimal performance is achieved with smaller LRs on the SD1, SD2, and Jasper ridge dataset, while on the Samson and Apex datasets, moderate LRs are advisable. This suggests that the sensitivity to LR varies across datasets. For WD, appropriate values help prevent overfitting. The optimal WD range is dataset-dependent: SD1 performs better with larger WD values, SD2 and Jasper ridge dataset prefers smaller values, and Samson and Apex datasets achieve the best results with moderate WD settings. Regarding the regularization coefficient β, different datasets show different preferences. Smaller β values yield better performance on the SD1, the SD2, and the Apex dataset, whereas a moderate β is more effective for the Samson dataset. For the Jasper ridge dataset, a smaller value of β is chosen. For γ, a larger value improves performance on the SD1, while smaller values are generally better for the other datasets. Overall, PGViT-TC-Net consistently outperforms PGViT-Net for all parameter settings. All experiments are conducted with a fixed number of training epochs (set to 200). Finally, the empirical set of hyperparameters for all datasets is presented in Table VIII.

D. Analysis of Different Decoder Weight Initialization Strategies

In this section, we analyze the effect of different decoder weight initialization strategies on the performance of PGViT-Net and PGViT-TC-Net. We evaluate both VCA-based and NMF-based initialization on all datasets, and the experimental results are reported in Table IX.

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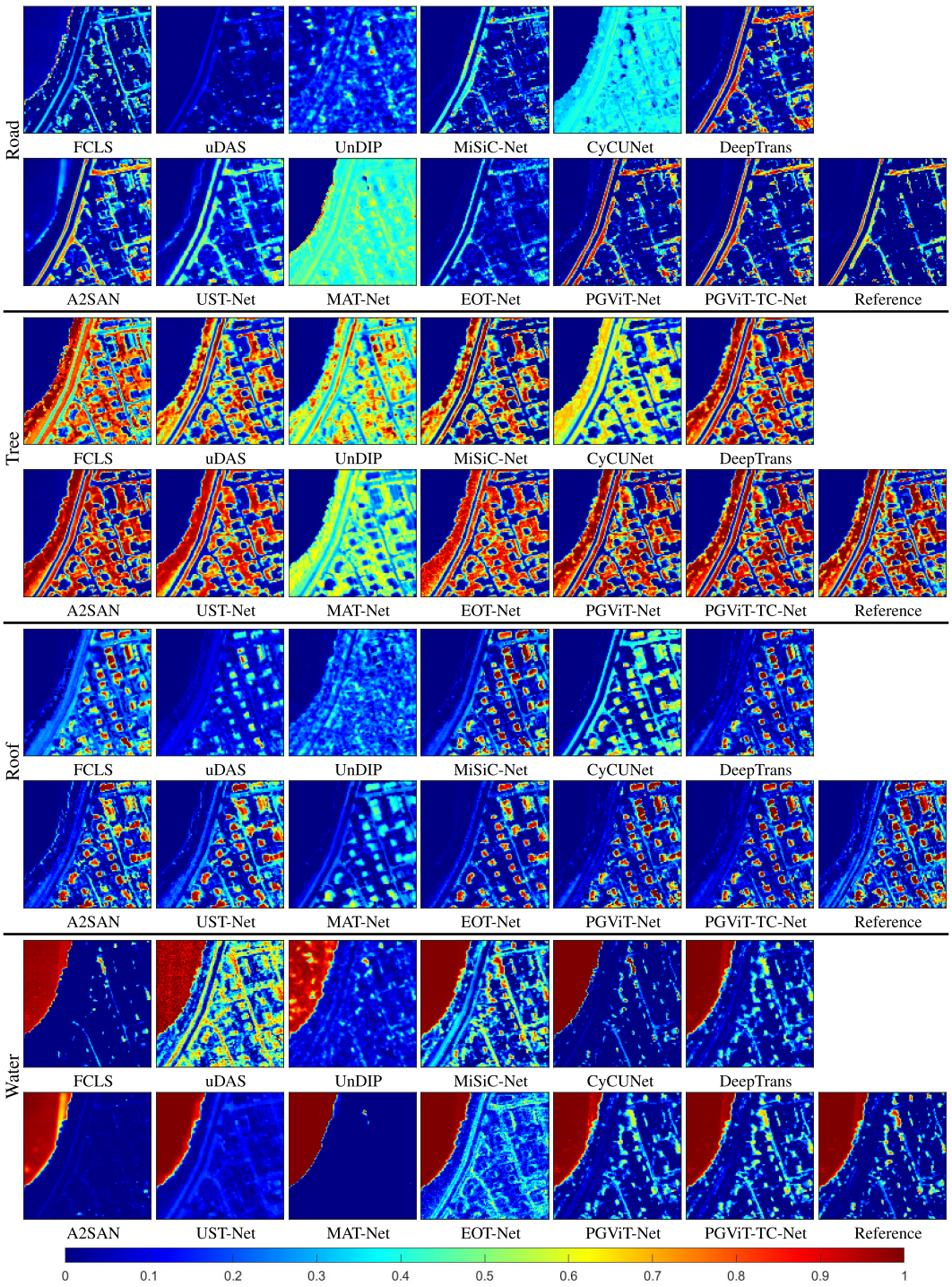


Fig. 13. References and estimated abundance maps of the Apex dataset using different methods.

For PGViT-Net, except for the RMSE on the SD1 dataset and the SAD on the SD2 dataset, where VCA-based initialization yields slightly inferior results compared to NMF-based initialization, VCA-based initialization achieves better performance in all other cases. For PGViT-TC-Net, VCA-based initialization performs slightly worse than NMF-based initialization on the SAD for the SD2 and Apex datasets, while outperforming NMF-based initialization on all remaining evaluation datasets. Overall, VCA-based initialization consistently achieves marginally better performance in most cases, while the overall performance gap between VCA-based and NMF-based initialization remains relatively small. This indicates that VCA primarily serves as a practical initialization strategy in the proposed method rather than a critical factor determining performance improvements. In general, the performance advantage of the proposed method mainly stems from the proposed PGE and its effective feature representation capability. For consistency in the experimental setup, VCA-based

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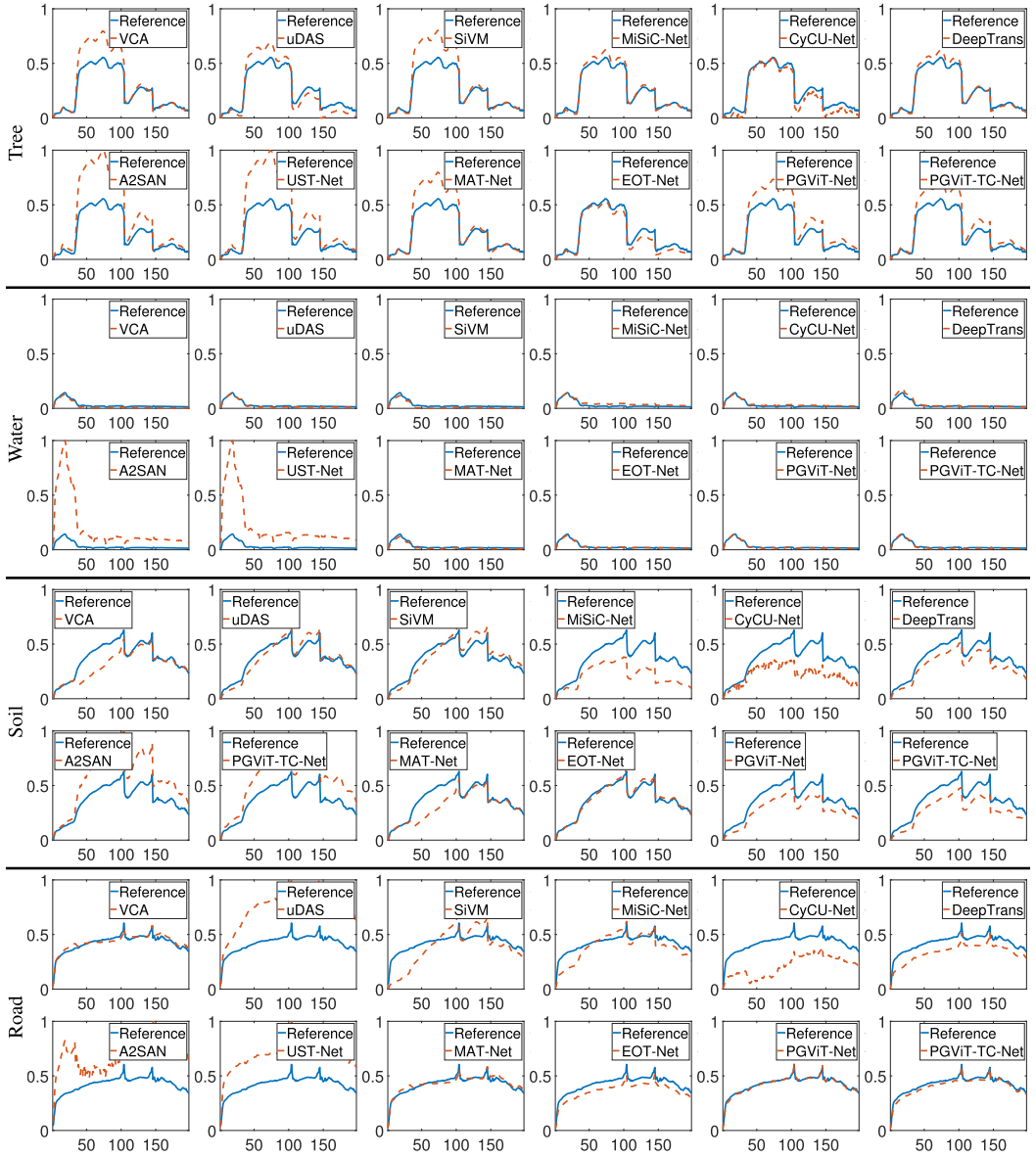


Fig. 14. Estimated endmember curves of the Jasper ridge dataset using different methods. The horizontal and vertical axes of each subfigure represent spectral bands and reflectance (%), respectively.

TABLE IX
IMPACT OF DIFFERENT DECODER WEIGHT INITIALIZATIONS. BEST RESULTS ARE IN BOLDFACE

Initialization		VCA		NMF	
		SAD	RMSE	SAD	RMSE
PGViT-Net	SD1	**0.10477**	0.07414	0.12963	**0.06583**
	SD2	0.00405	**0.05628**	**0.00399**	0.05730
	Samson dataset	**0.03947**	**0.04301**	0.06607	0.04752
	Apex dataset	**0.09906**	**0.10183**	0.09939	0.10365
	Jasper ridge dataset	**0.03720**	**0.06640**	0.06588	0.10932
PGViT-TC-Net	SD1	**0.09104**	**0.05251**	0.12517	0.07205
	SD2	0.00408	**0.05642**	**0.00394**	0.05750
	Samson dataset	**0.03718**	**0.04009**	0.06599	0.04731
	Apex dataset	0.09607	**0.09571**	**0.08858**	0.11906
	Jasper ridge dataset	**0.05330**	**0.06535**	0.08162	0.10758

initialization is adopted for the decoder weights throughout this work.

E. Ablation of the Proposed Method on Different Datasets

In this section, we analyze the ablation experiments of the proposed PGViT-Net, with a particular focus on the PGE module. Table X reports the ablation results of the PGE module on all datasets. In particular, the experiment systematically investigates the impact of key components on model performance by individually enabling or disabling the CNN_1 and transformer modules.

As shown in Table X, the model achieves the best performance across all datasets when both CNN_1 and the transformer are jointly employed, yielding the lowest SAD and RMSE values. When either module is removed, the performance degrades to varying degrees, indicating that CNN_1 plays a crucial role in local feature extraction and spatial modeling, while the transformer is essential for capturing long-range dependencies and global contextual information. When both modules are removed, the performance deteriorates further, which validates the rationality and effectiveness of the pro-

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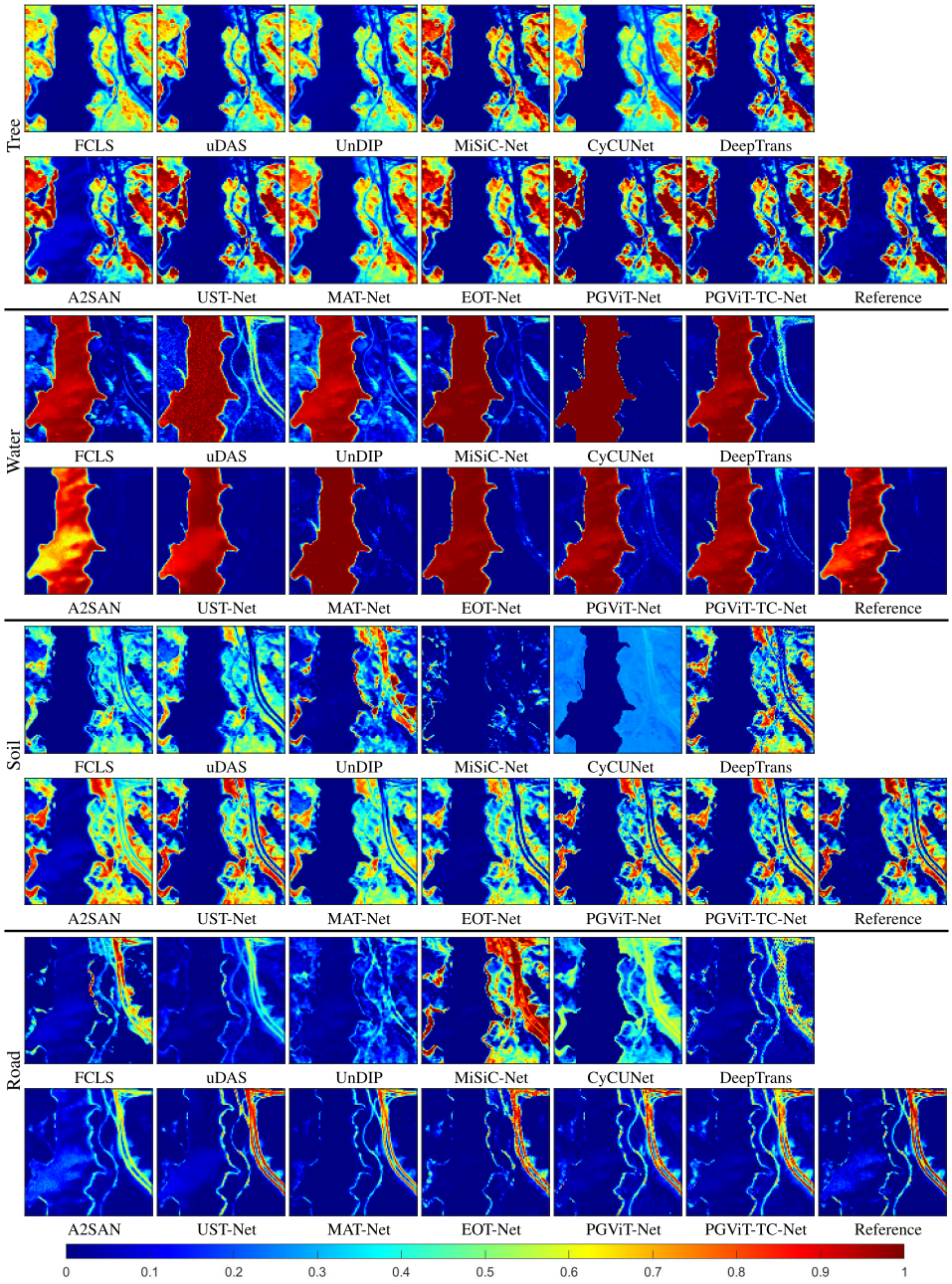


Fig. 15. References and estimated abundance maps of the Jasper ridge dataset using different methods.

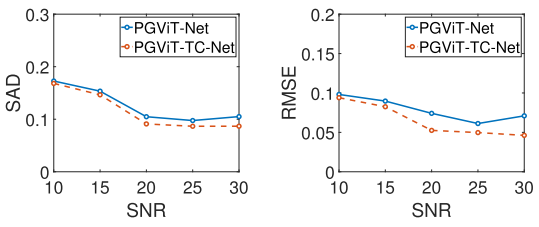


Fig. 16. SAD and RMSE on the SD1 under different noise levels.

posed network design. Meanwhile, based on the performance results on SD1 and the Apex dataset, when both modules are disabled, the results are better than when either module is disabled individually, indicating a synergistic interaction between CNN_1 and the transformer. These results clearly demonstrate the importance of jointly modeling convolutional structures and transformers in PGE for improving HU performance.

F. Running Time of All Methods on Different Datasets

In this section, we report the running time of all compared methods on different datasets, as summarized in Table XI. It can be observed that traditional optimization-based methods (e.g., FCLS), owing to their simple model structures, exhibit the lowest computational cost across all datasets. In contrast, UnDIP and UST-Net generally incur higher running times, and the computational burden varies notably among different methods depending on their architectural designs. In particular,

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Fig. 17. SAD and RMSE on the different datasets exploiting different parameters.

TABLE X

ABLATION STUDY OF PGE ON DIFFERENT DATASETS. BEST RESULTS ARE IN BOLDFACE

PGE		SD1		SD2		Samson dataset		Apex dataset		Jasper ridge dataset	
CNN_1	transformer	SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE
✓	✓	**0.10477**	**0.07414**	**0.00405**	**0.05628**	**0.03947**	**0.04301**	**0.09906**	**0.10183**	**0.03720**	**0.06640**
×	✓	0.11195	0.08285	0.00406	0.05757	0.25080	0.29496	0.11629	0.12676	0.05522	0.09708
✓	×	0.11274	0.07903	0.00457	0.05660	0.06649	0.09673	0.29725	0.18871	0.12330	0.09696
×	×	0.10976	0.07854	0.00925	0.08005	0.31299	0.38276	0.14027	0.11564	0.12745	0.09374

TABLE XI

RUNNING TIME OF ALL METHODS ON DIFFERENT DATASETS (IN SECONDS). BEST RESULTS ARE IN BOLDFACE

Algorithms	FCLS	uDAS	UnDIP	MiSiC-Net	CyCU-Net	DeepTrans	A2SAN	UST-Net	MAT-Net	EOT-Net	PGViT-Net	PGViT-TC-Net
SD1	**0.67**	6.33	146.89	38.76	15.88	12.21	21.27	262.16	13.69	58.30	13.87	14.84
SD2	**0.20**	5.67	202.16	41.58	20.53	17.56	35.63	256.66	18.12	39.94	18.92	19.97
Samson dataset	**0.35**	6.06	226.12	47.58	26.57	23.89	31.48	251.74	24.77	44.74	26.15	26.88
Apex dataset	**1.19**	262.79	516.50	60.17	31.58	31.43	40.09	464.35	32.18	60.71	39.04	39.73
Jasper ridge dataset	**0.83**	136.65	312.73	50.92	20.99	28.51	35.06	509.23	26.78	54.68	30.64	31.26

MiSiC-Net, CyCU-Net, DeepTrans, A2SAN, MAT-Net, and EOT-Net present different levels of computational overhead on different datasets. It is worth noting that the proposed PGViT-Net and PGViT-TC-Net achieve relatively stable

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running times that are comparable to those of DeepTrans and MAT-Net.

V. Conclusion

In this article, we propose a novel encoder module, PGE, primarily guided by the physical mechanisms of unmixing during the encoding process. Based on this encoder, we introduced a new unmixing network, called PGViT-Net, which does not require additional modifications to the data shape, making it more aligned with the unmixing process. Furthermore, we proposed PGViT-TC-Net to enhance the utilization of spatial–spectral and local–global information. The proposed approaches demonstrated their validity against some state-of-the-art HU approaches on two synthetic and three real datasets. In the future, potential exploration may consider tokens generated from proportion maps corresponding to spectral bands that may include second-order reflections in each hyperspectral band image for long-range feature modeling.

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